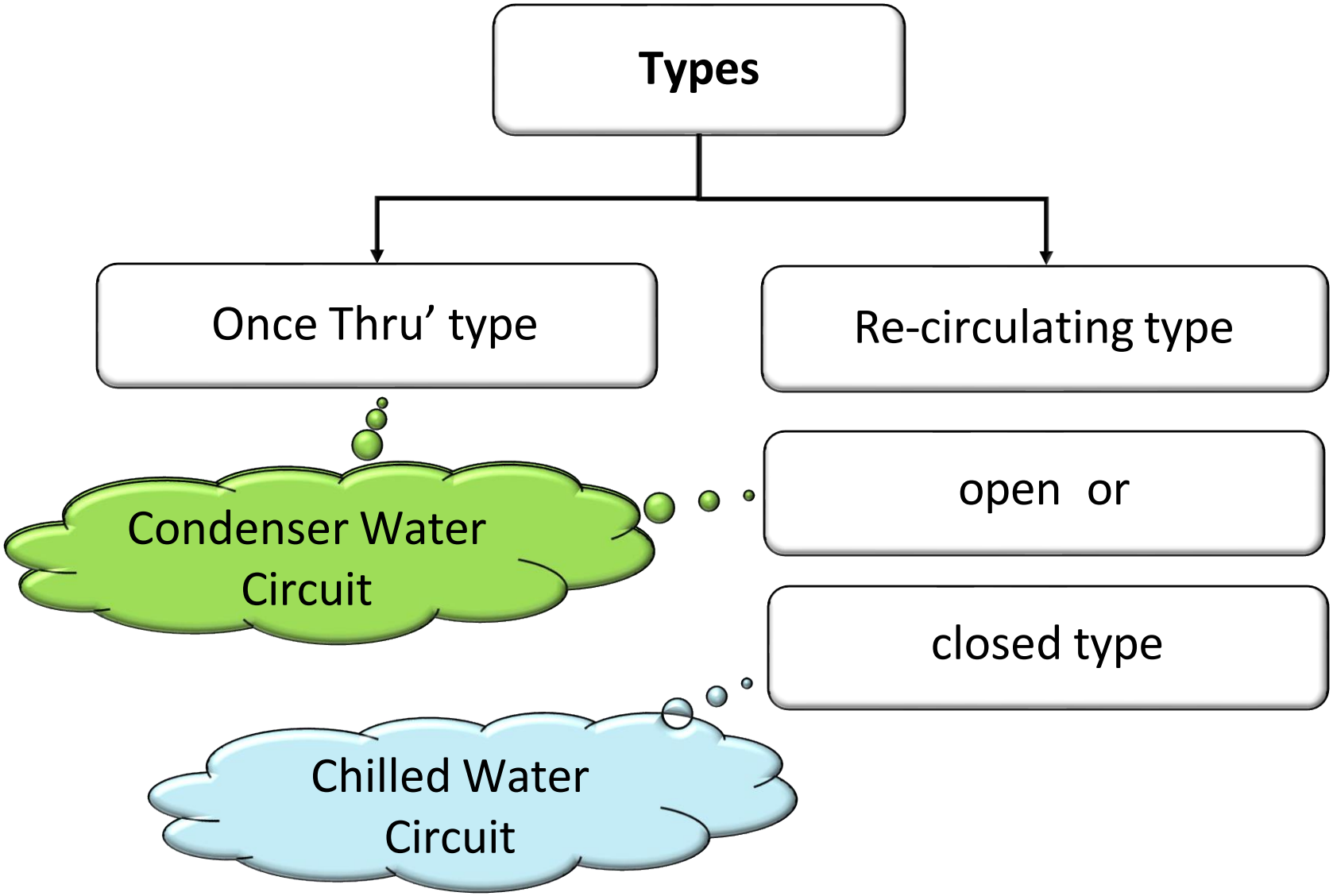
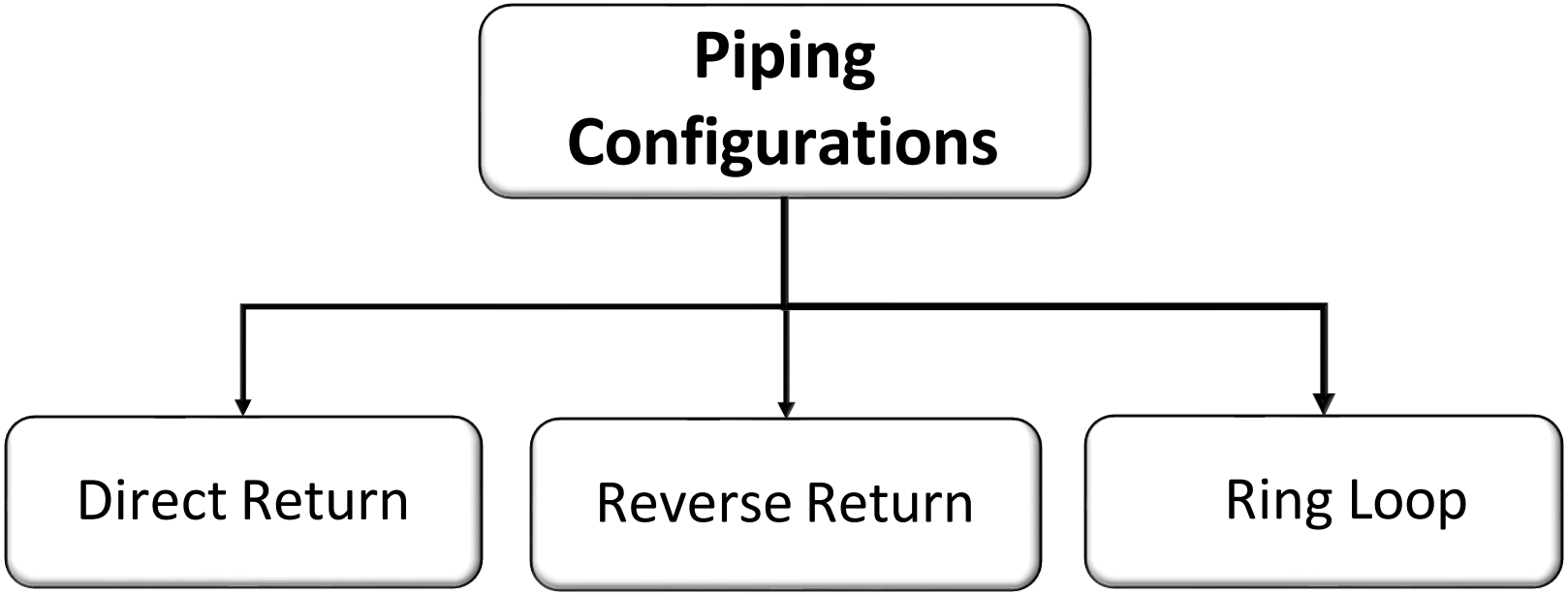
Lesson 6 & 8

PIPE SYSTEM & SIZING

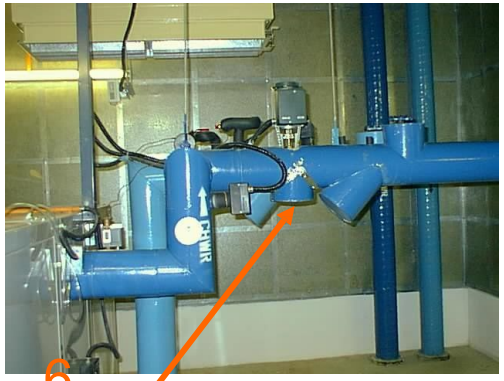
Types of Water Circuits

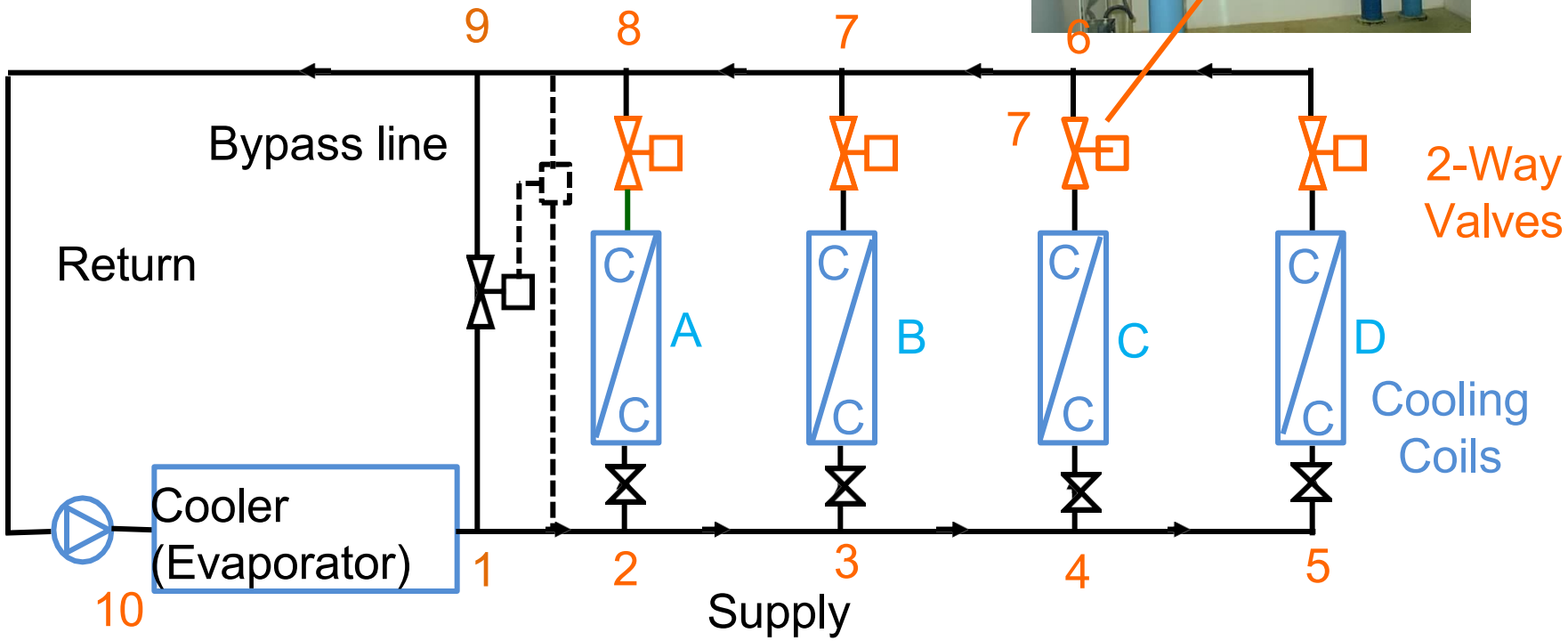


Chilled Water Circuits

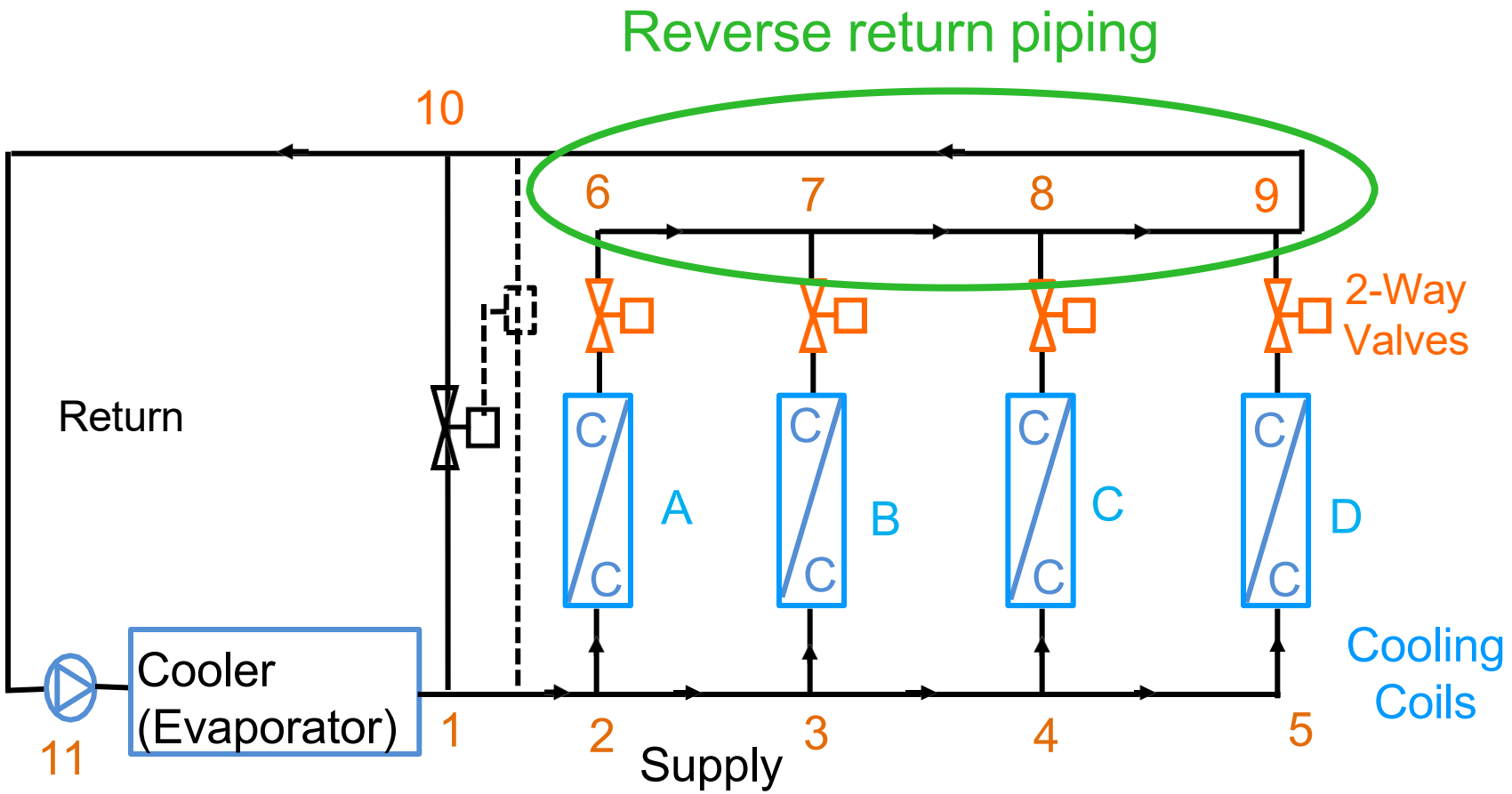


Direct Return

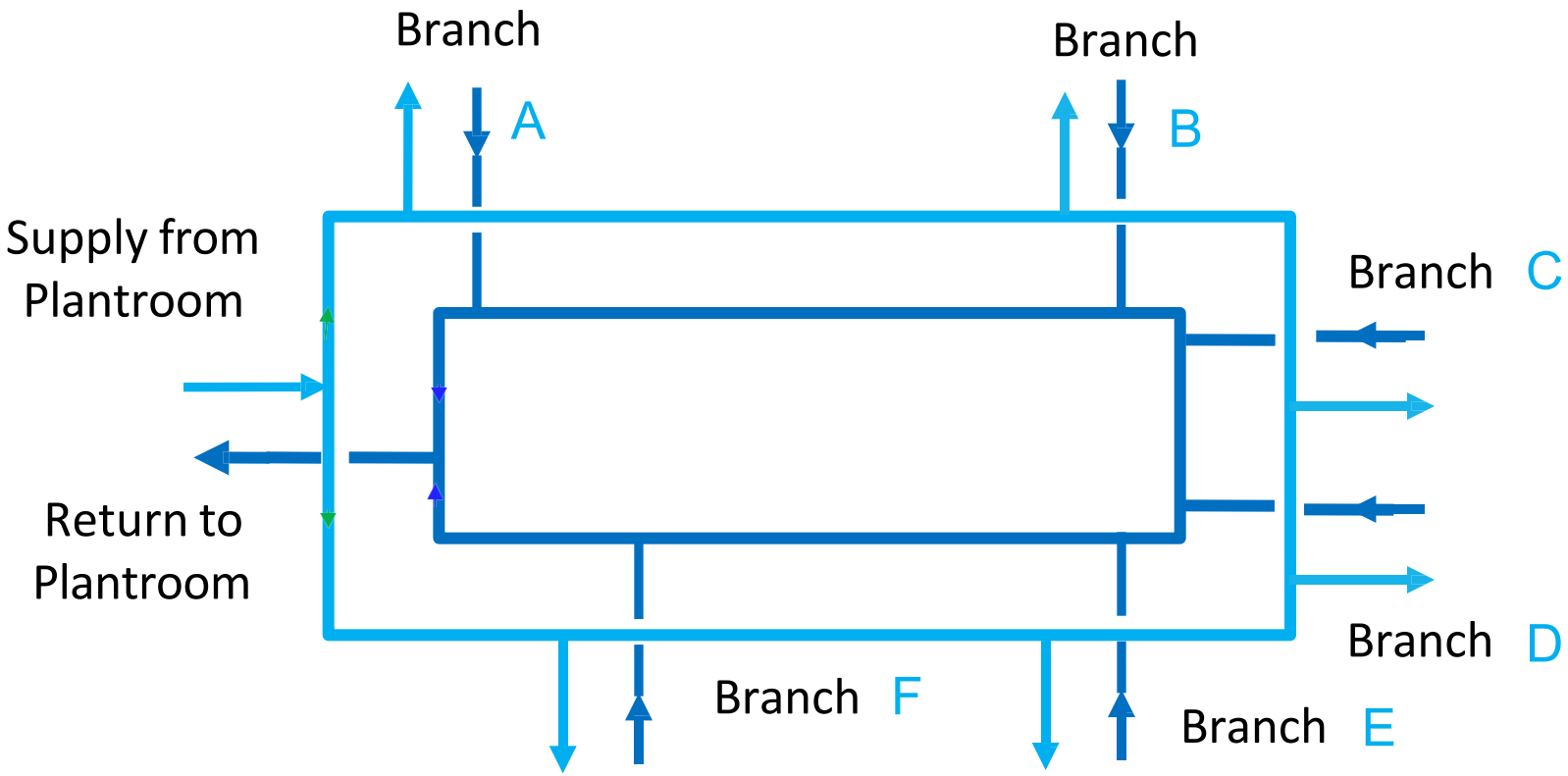




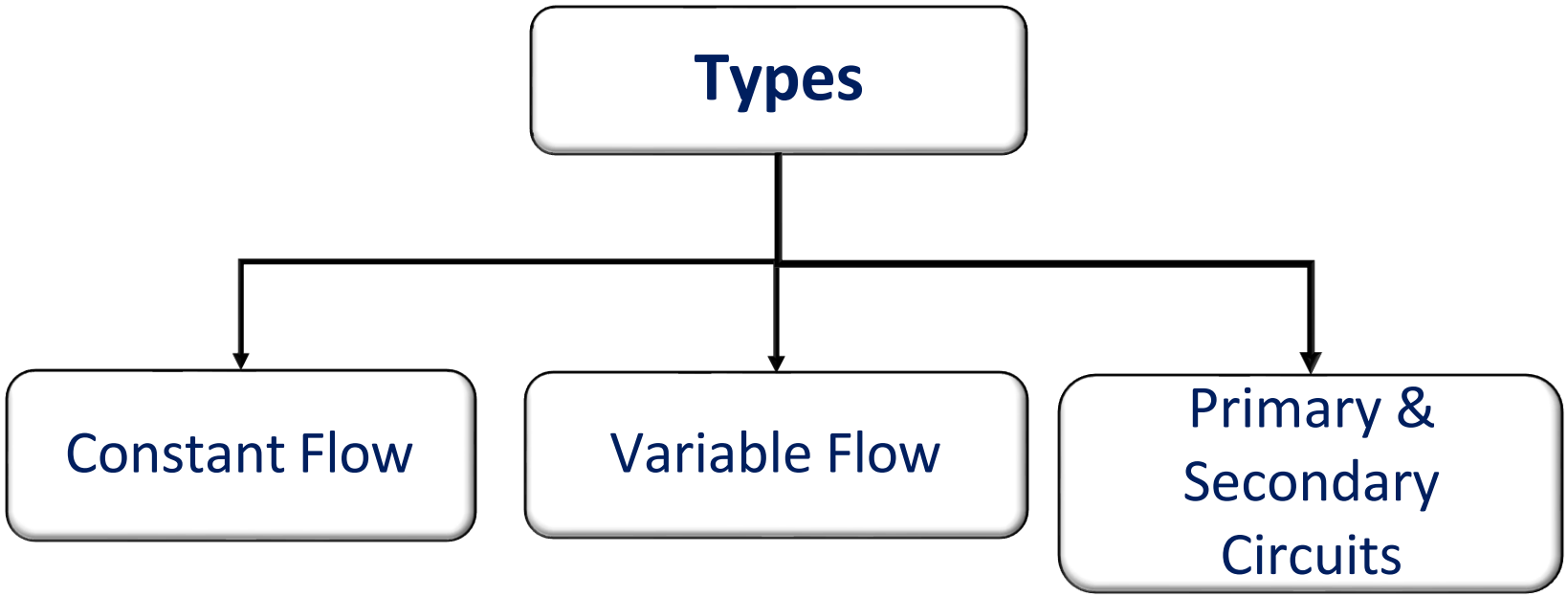
Reverse Return



Loop System


Branch
A
Branch
B
Supply from
Plantroom
Branch C
Return to
Plantroom
Branch D
Branch F
Branch E

Types of Chilled Water Circuits



Constant Flow



- Chilled water pumps run at a constant speed.
- Reduction in demand matched by diverting chilled water through three-way modulating valves.
- Flow rate in main circuit remains constant.

Variable Flow

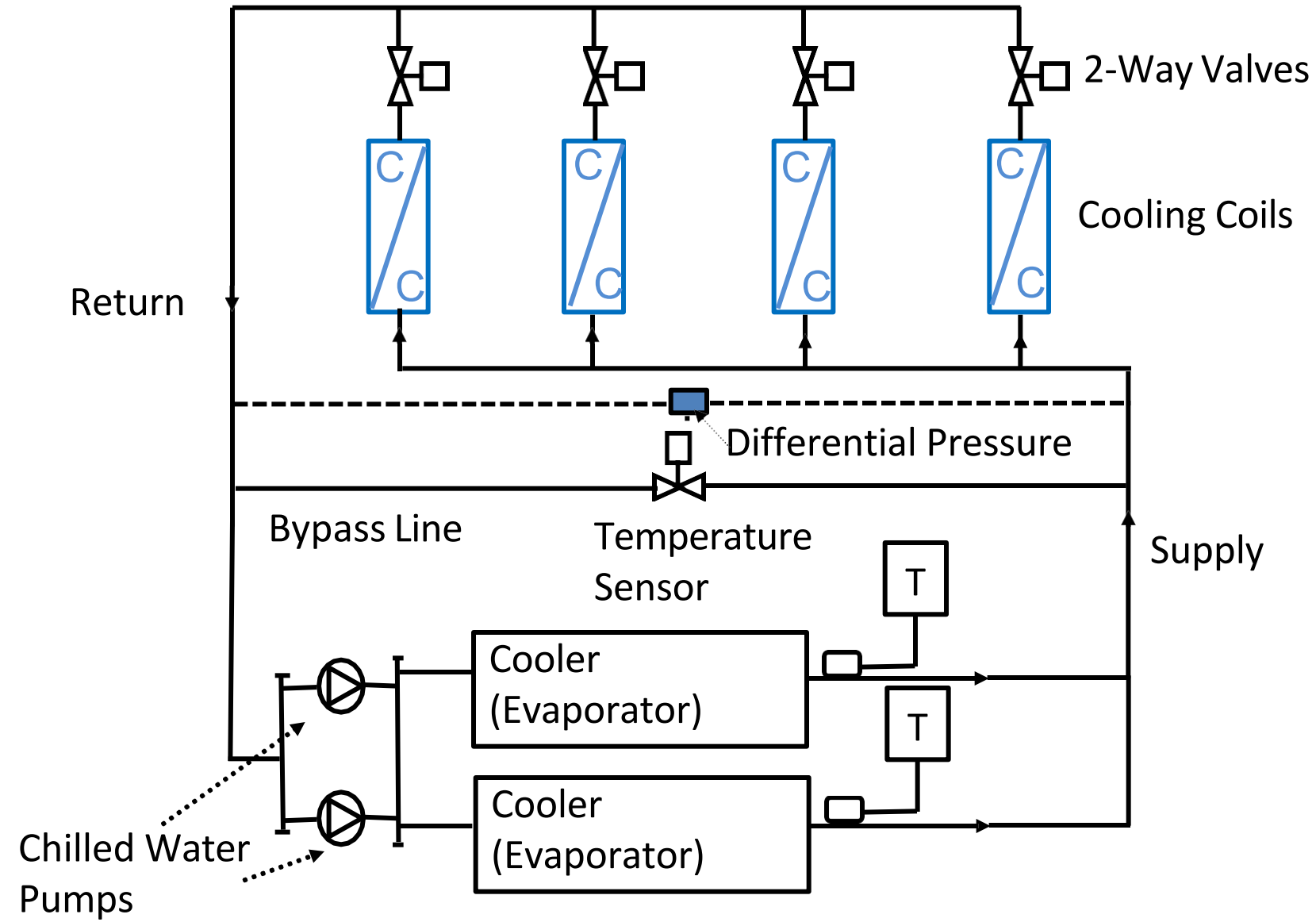
Constant Flow Chillers

- Reduction demand, two-way modulating valve restrict amount of chilled water.
- Pressure bypass line to bypass water back to chillers
- Flow rate in main circuit remains constant.
- If bypass valve fully opened, one chiller and chilled will shutdown.

Variable Flow Chillers

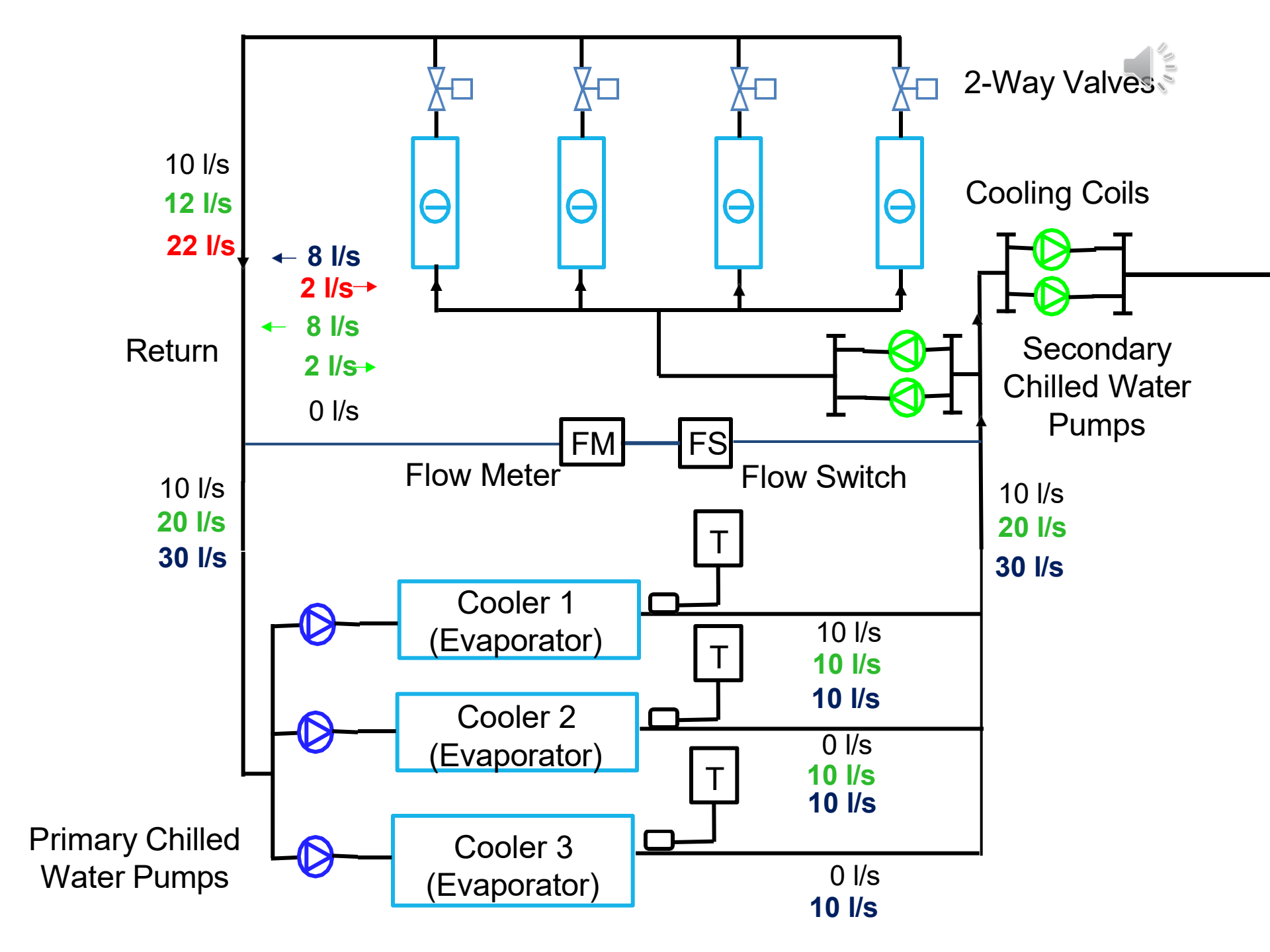
- Chillers installed are suitable for variable flow through the chillers
  - VSD chilled water pumps
  - pressure bypass control required for low load conditions.

Two Chillers with 2-way Valves and Pressure Bypass



Primary & Secondary Circuits

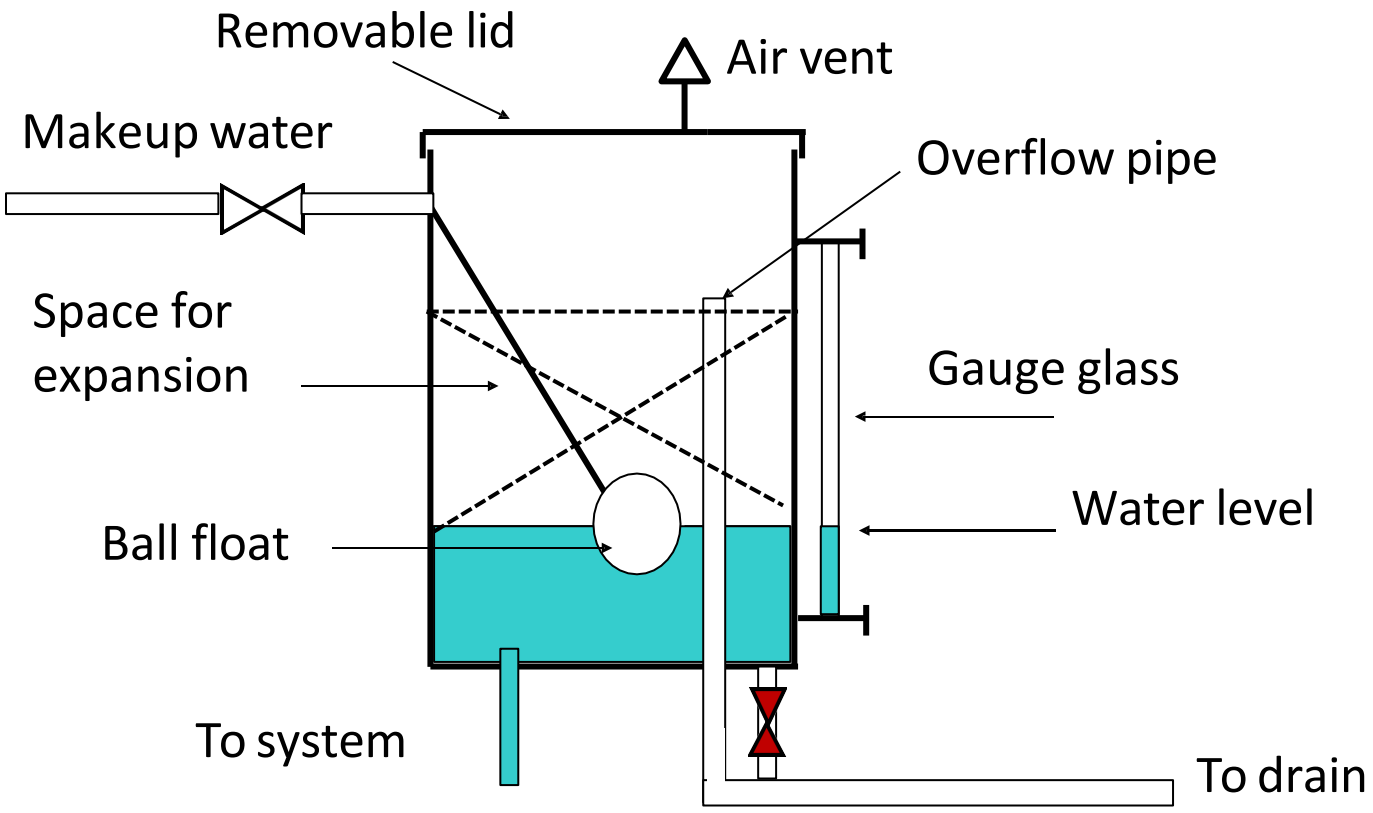
- Separate set of pumps for chillers - "chiller or primary loop"
- Separate set of pumps for distribution system - "distribution or secondary loop".
- Variable speed pumps can be used for both primary and secondary circuits


2-Way Valves
Cooling Coils
10 l/s
12 l/s
22 l/s
8 l/s
2 l/s
8 l/s
2 l/s
0 l/s
Return
Secondary Chilled Water Pumps
FM
FS
Flow Meter
Flow Switch
10 l/s
20 l/s
30 l/s
10 l/s
20 l/s
30 l/s
T
Cooler 1 (Evaporator)
10 l/s
10 l/s
10 l/s
T
Cooler 2 (Evaporator)
0 l/s
10 l/s
10 l/s
T
Cooler 3 (Evaporator)
0 l/s
10 l/s
Primary Chilled Water Pumps

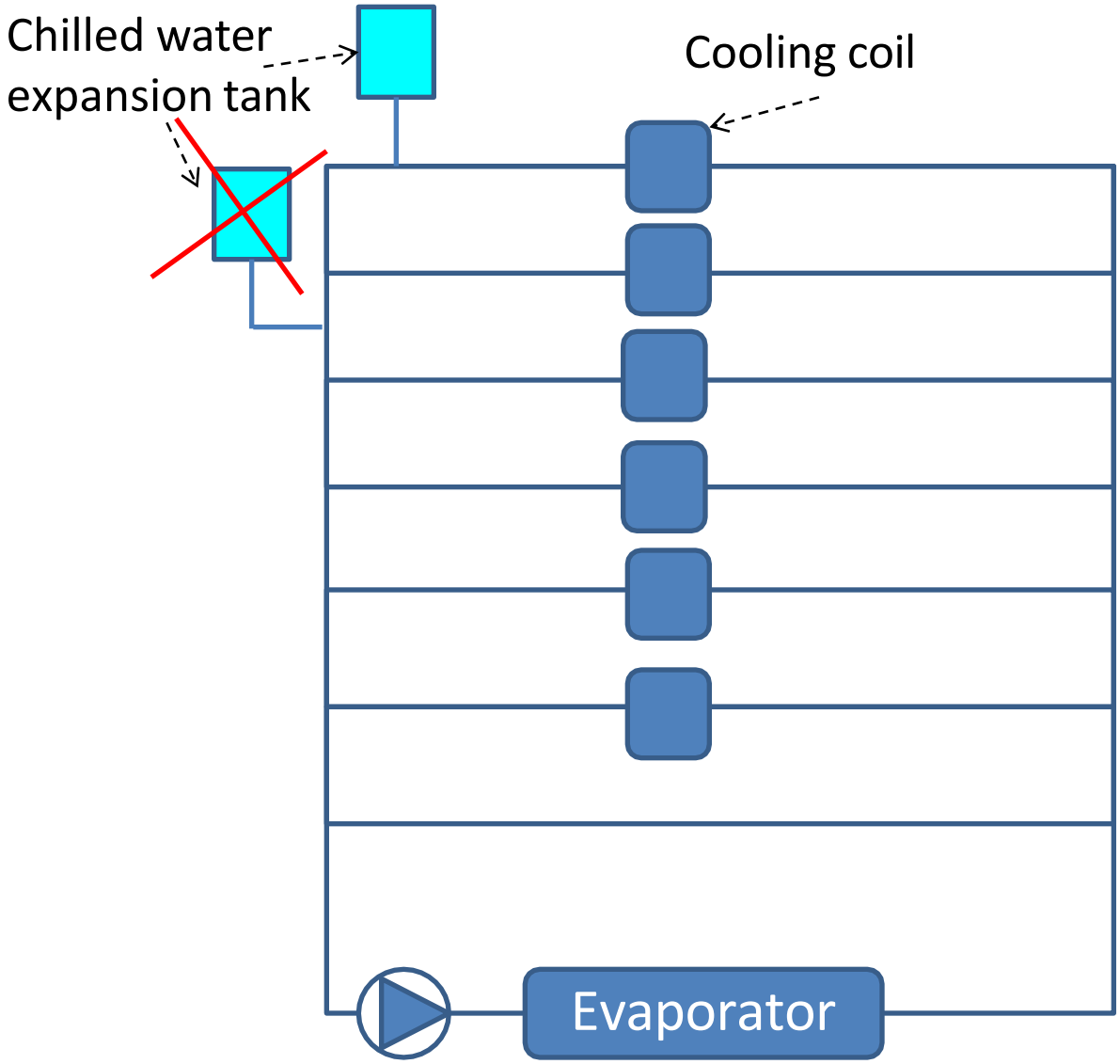
Chilled Water Expansion Tank

- Functions
  - Allow for the expansion of water
  - Point for adding water
- Types
  - Open type
  - Closed type (diaphragm)

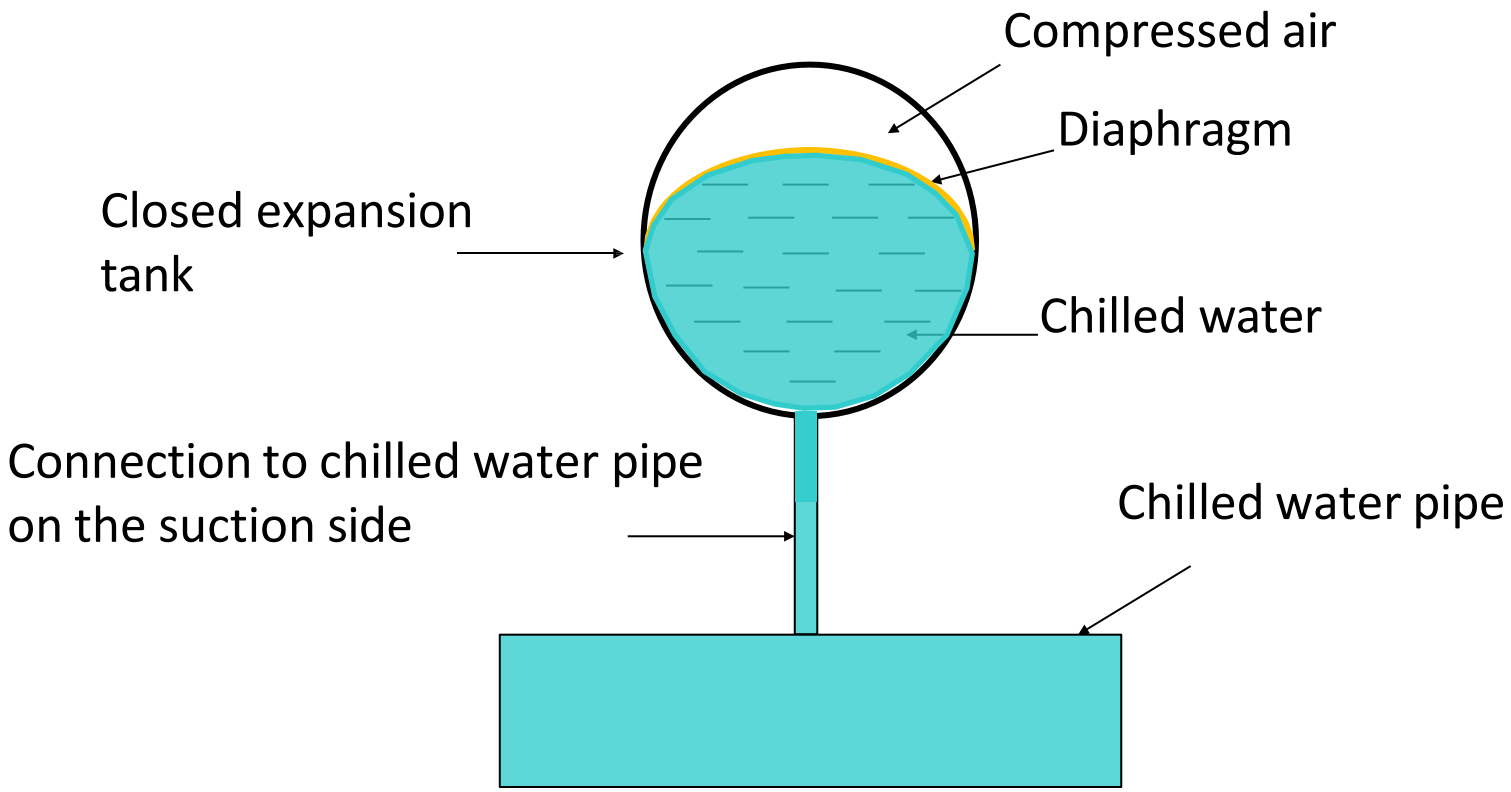
Open Chilled Water Expansion Tank



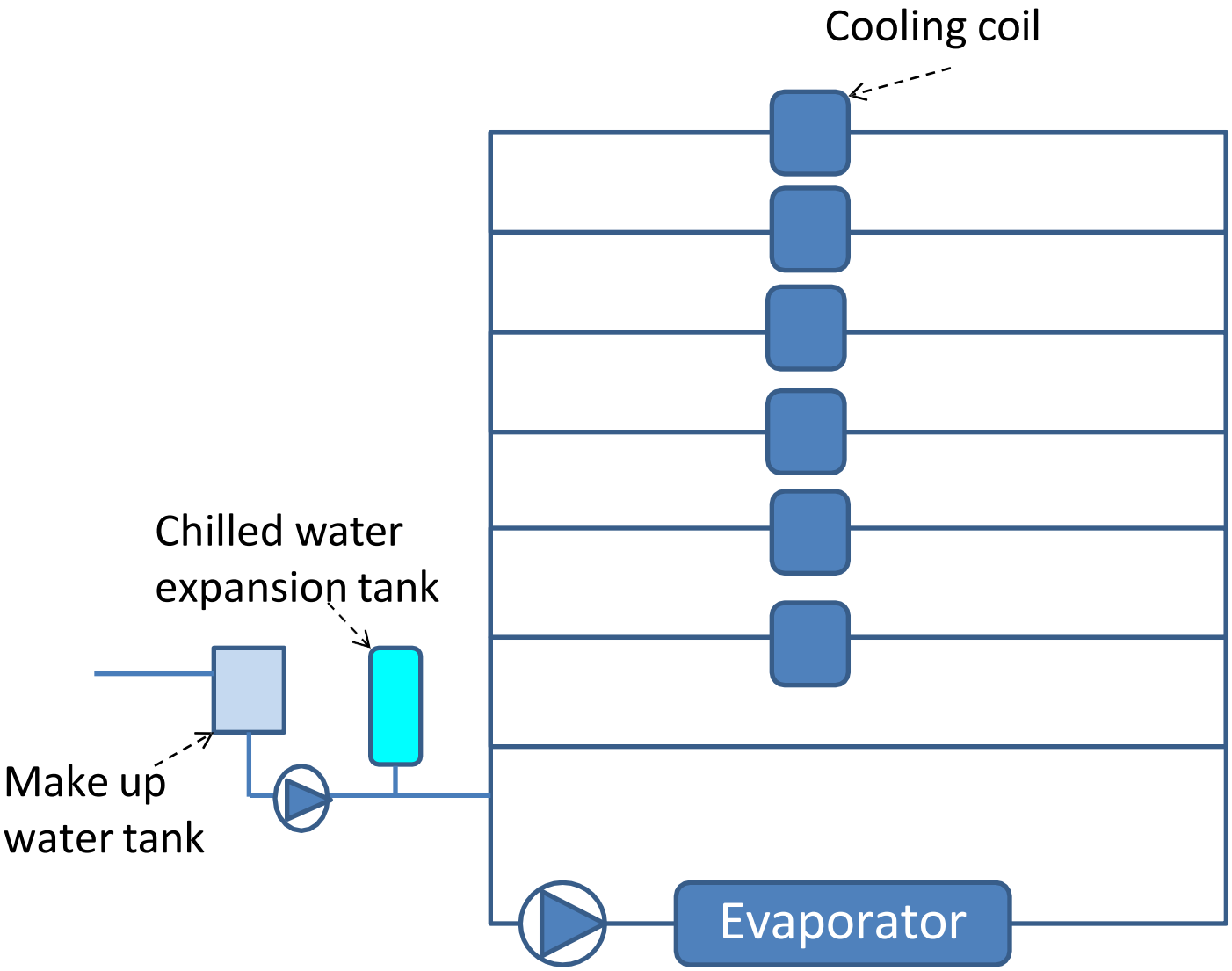
Open Chilled Water Expansion Tank



Closed Chilled Water Expansion Tank



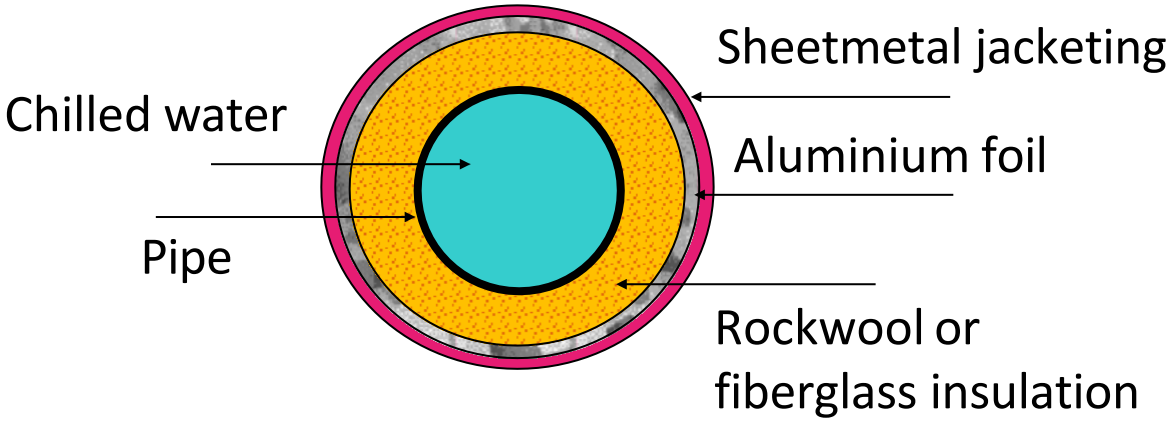
Closed Chilled Water Expansion Tank



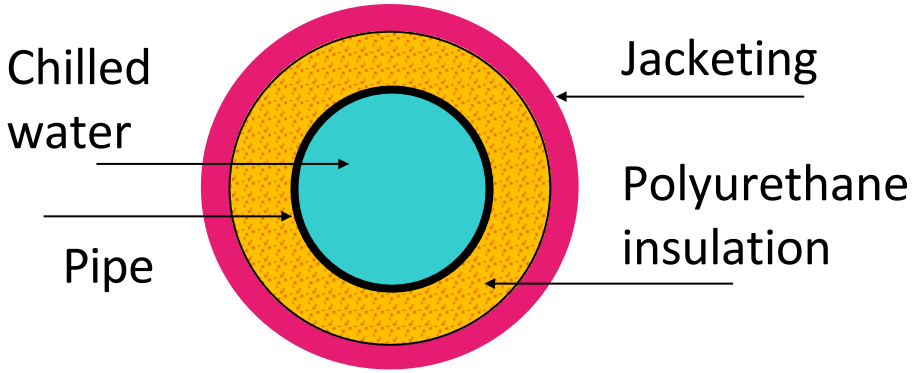
Pipe Insulation

- Water at temperatures (5°C to 12°C) - below the dewpoint of the surrounding air
- Insulation
  - Pre-insulated
  - On-site insulation

Pipe Insulation

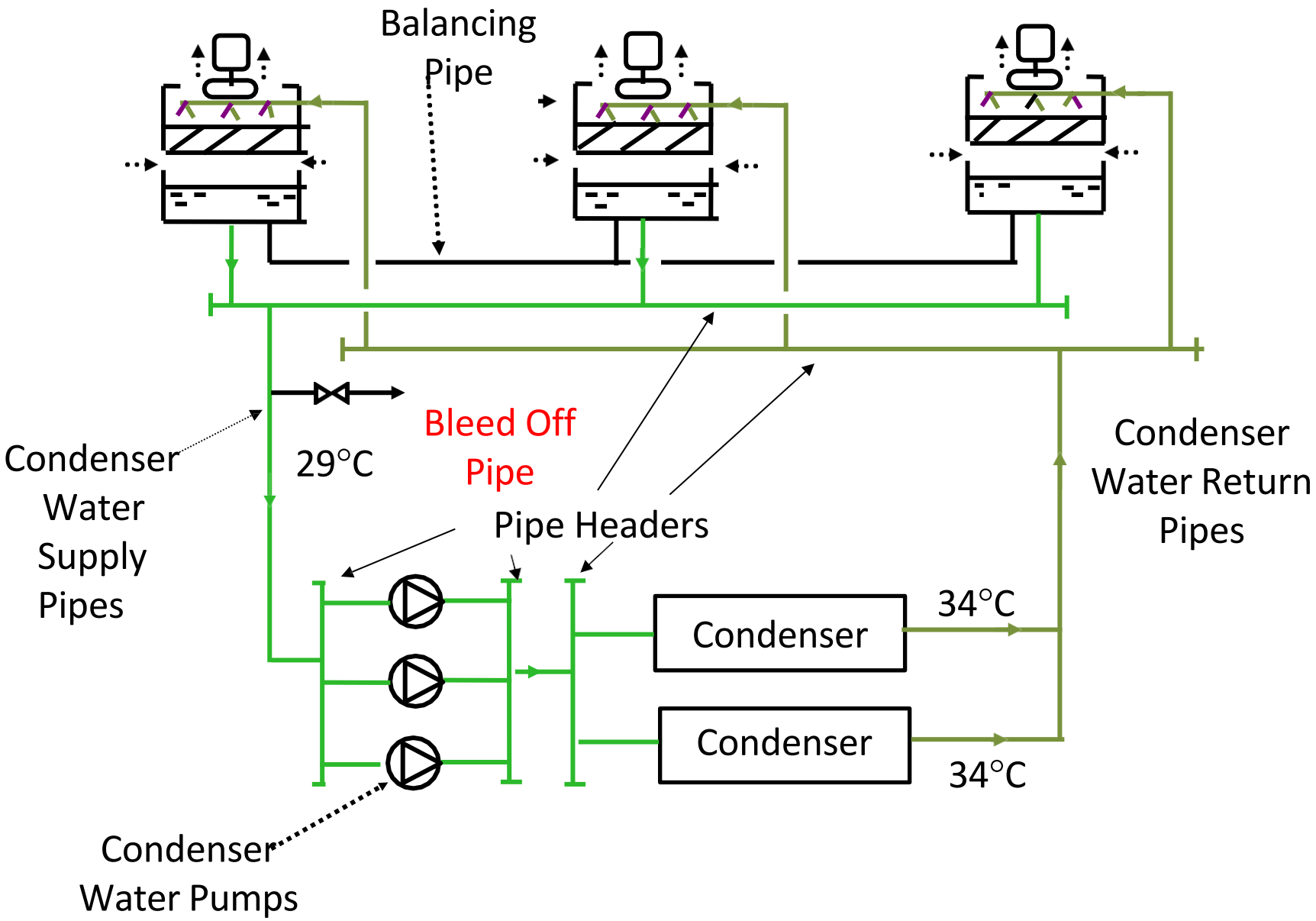


On-site Insulation



Pre-insulated Pipe

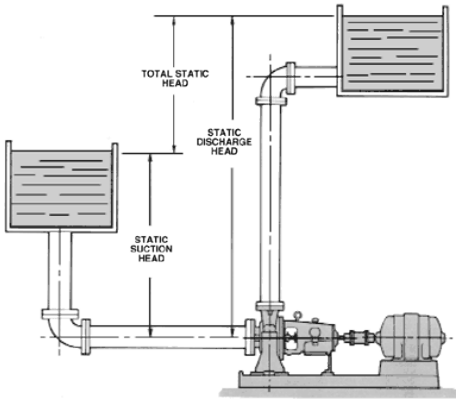
Condenser Water Circuit

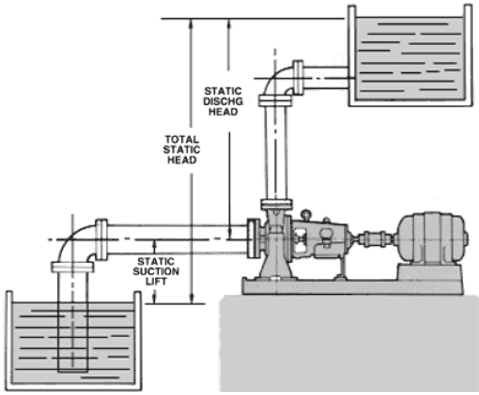

Balancing
Pipe
Condenser
Water
Supply
Pipes
29°C
Bleed Off
Pipe
Pipe Headers
Condenser
Water Return
Pipes
Condenser
34°C
Condenser
34°C
Condenser
Water Pumps

Principles for Design Circuits

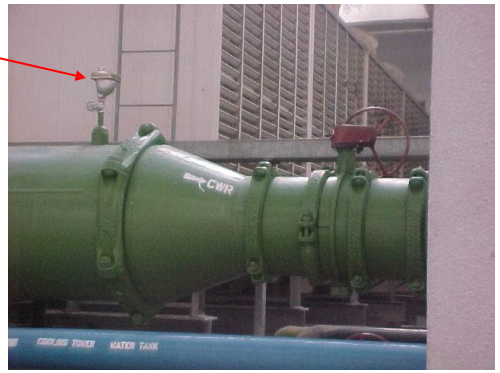
- Nine important principles

- Centrifugal pumps must always be primed.
- Cavitation must be avoided.
- Air must not be drawn into system.
- All air must be vented from system.
- In open circuits, as much of pipework as possible must be below static water level.





Air vent



Principles for Design Circuits

- All control valves must be below static water level in open circuits.
- Attention must be paid to positive head.
  - Use open vents used when there is risk of warm water boiling and must not to be located to discharge continuously into feed and expansion tank.
  - Connection from feed and expansion tank must be made at a place such that principles are followed.

Pipe Sizing

- Chilled Water lines should be selected for optimum size with respect to:-
  - Initial cost
  - Operation and maintenance cost
  - Water velocity
  - Pressure drop

Pipe Sizing

- Desirable to have line sizes as small as possible based on low initial cost.
  - Small pipe sizes ➔ high water velocity and pressure drop.
  - High water velocity ➔noise problems, rapid erosion and early failure of pipe.
  - High pressure drop ➔increase pump operating cost and may require a larger size pump

Water Velocities Permissible

- Maximum velocity permissible is established by:-
  - Noise generated by water flowing through pipe.
  - Erosion of piping by water and entrained sand, air and other foreign particles.

Water Velocities Permissible

- Minimum water velocity permissible established by:-
  - need to keep any air, which tends to collect in certain parts of the system, moving
  - Need to prevent sediments from settling down in the pipe sections.

Friction Loss

- Once through systems must be sized to provide required flow at a pressure loss within pressure available after all other losses (condenser pressure drop, hydrostatic, valve and fitting losses) have been deducted.
- Re-circulating pump systems are sized to provide reasonable balance between increased pumping power due to high friction loss and increased piping cost due to larger pipe sizes.

Allowance for Aging

- With age, internal surfaces of pipes become increasingly rough, which reduces the available pressure with a fixed pump pressure. However, designing with excessive age allowances may result in oversized piping.
- Allowances for expected decreases in capacity are sometimes treated as a specific amount (%).

Sizing Method – Equal Friction

- Prepare a sketch of piping system.
- Find flow rate in each section of pipe by adding up required flow rates for each terminal unit, working back from last unit to the main from the pump.
- Choose a value of friction loss rate for the main from the pump, based on the following limits:-

Sizing Method – Equal Friction

- Friction loss rate – between 100 - 400 Pa/m, with an average of 250 Pa/m.
- Recommended water velocities:

Diameter (mm)	Velocity Range (m/s)
125 mm and above	2.0 – 3.6
50 – 100 mm	1.0 – 2.0
25 mm and below	0.5 – 1.0

Sizing Method – Equal Friction

- Select pipe size for main pipe from pump, from friction loss chart based on limits. If there is more than one choice, select smaller pipe size if initial costs are more critical or larger size if operating costs are more critical.

- Continuing along circuit chosen, select succeeding pipe sizes according to the following guides.

Sizing Method – Equal Friction

- When flowrate decreases significantly, select next smaller pipe size from chart so that friction loss rate is approximately equal to value chosen for initial section (equal friction).
- Do not let velocity decrease below 0.6 m/s, to prevent the settling of dirt and air in the lines.
- Select pipe sizes for branches in other circuits. If a circuit is not extremely short, use the same friction loss rate. For a very short circuit, this procedure would result in a total pressure drop much less than that for the longest circuit, and cause problems of balancing flow.

Sizing Method – Equal Friction

- Therefore, select smaller pipe sizes in these branches to create increased pressure drop. This problem does not arise in a reverse return type system.
- Determine by inspection which circuit will be the longest or have the greatest equivalent length.
- Calculate pressure drop in this circuit.

Chilled Water Piping

- Prepare chilled water system schematic drawing.
- Accumulated quantities of chilled water flow in each pipe, calculated and shown on schematic drawing.
- Like ducting system, sizing and estimation of head losses in index run are then carried out.
- Head loss in index run is used together with total chilled water flow rates for selection of chilled water pump(s).

Condenser Water Piping

- Procedure similar to that of chilled water circuit, except that equipment in circuit comprises condenser of refrigeration machine, condenser water pumps and cooling towers.

SS 553 : 2016 – Pumping System Design Criteria

- Clause 10.7 of SS 553:2016 stipulates the pumping system design criteria.
- Sub-clause 10.7.1 on Hydronic system design and control states that the air-conditioning hydronic systems having a total pump system power exceeding 7.5 kW shall comply with 10.7.2 to 10.7.5
- Pump power limitation for chilled water systems shall be 349 $kW/m^3/s$.
- Pump power limitation for condensing water systems is 301 $kW/m^3/s$.

SS 553 : 2016 – Pumping System Design Criteria

- Motors exceeding 3.7 kW shall have controls and/or devices (such as variable speed control) that will result in pump motor demand of no more than 30% of design wattage at 50% of design water flow.

- Section 16 of the SS 553:2016 covers the general design requirements and installation of pipework.

Friction Loss Chart for Water in Commercial Steel

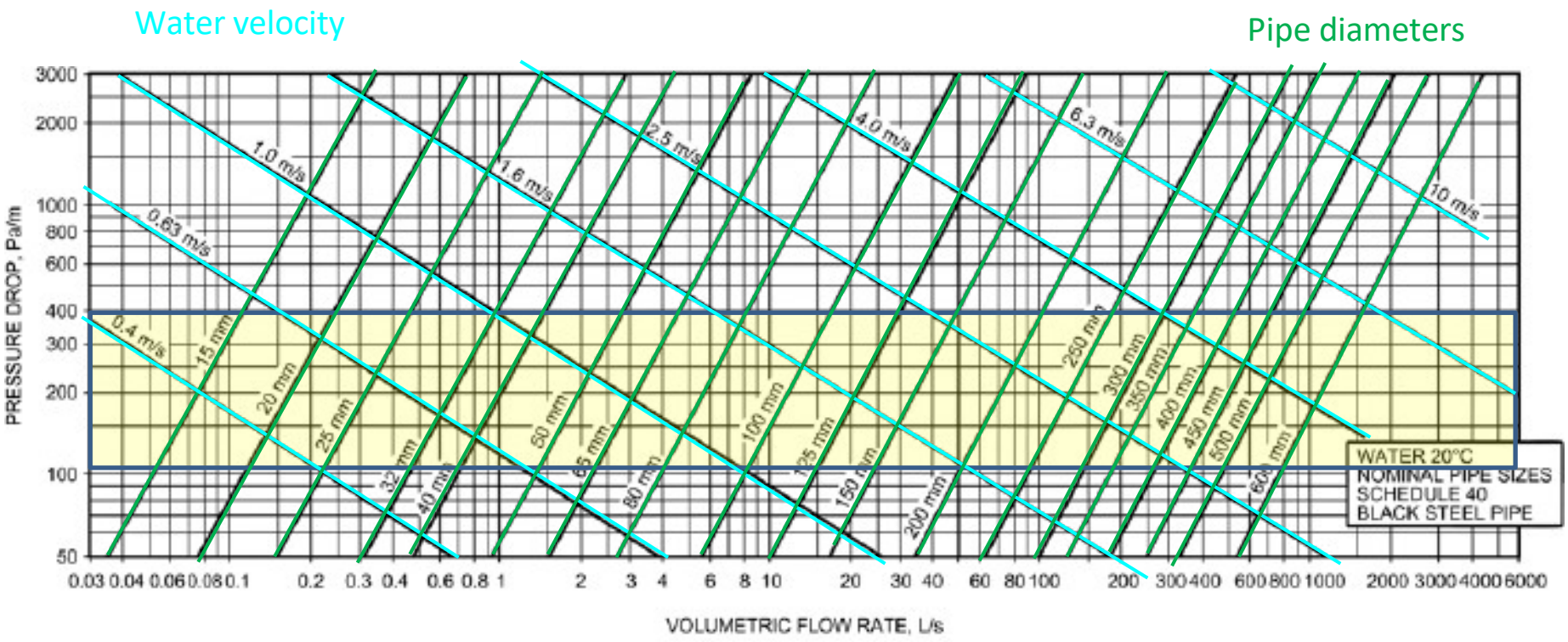
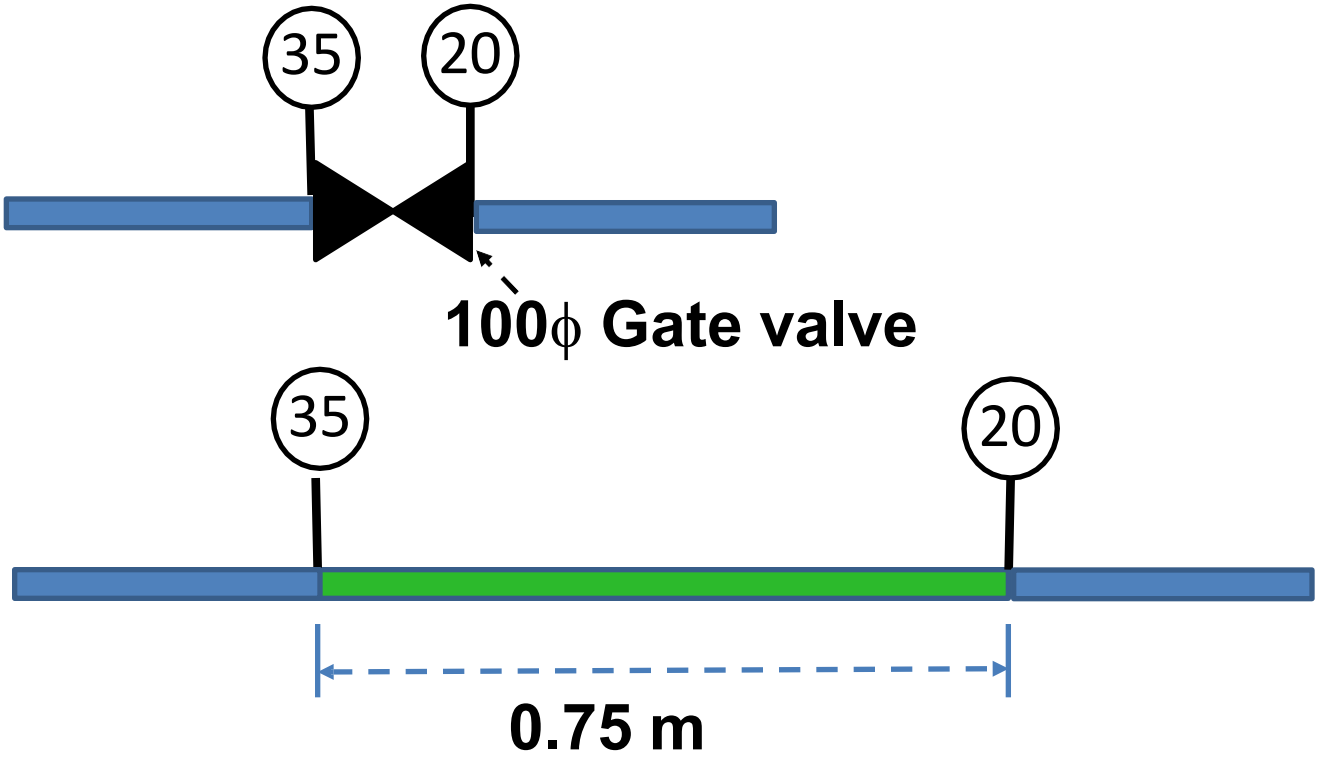


Fig. 4 Friction Loss for Water in Commercial Steel Pipe (Schedule 40)

Equivalent Length of Pipe Fittings and Valves

Equivalent Length of a pipe fitting - length of pipe of same size as fitting that would give rise to same pressure drop as fitting.

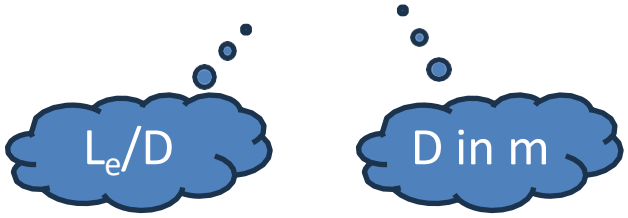


Equivalent Length of Pipe Fittings and Valves

For a given type of fitting the Equivalent Length (L_e) divided by inside diameter of pipe (D) i.e. (L_e/D) obtained is virtually constant for that type of fitting.

Figure 6.11 (b) shows L_e/D for various fittings.

L_e for 100mm globe valve = 320 x 0.1 = 32 m



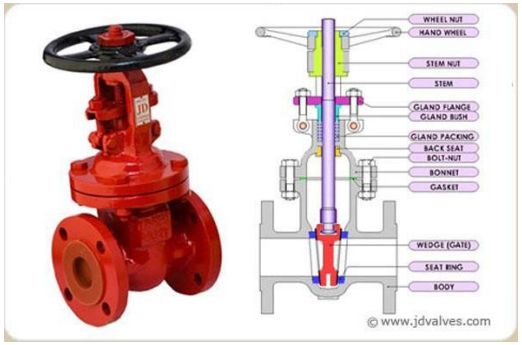
(Le/D) Ratio for Fittings in Steel Pipe

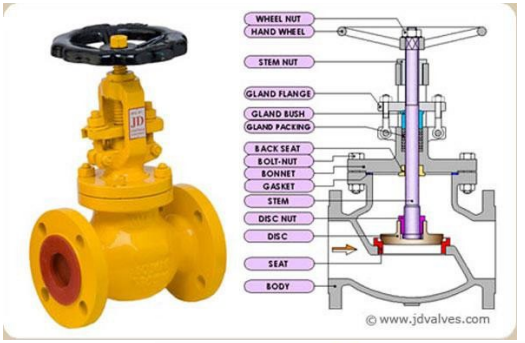
Fittings	L_e/D
Threaded Elbows	
90°, r/d=1	30
45°, r/d=1	16
Welded Elbows	
90°, sharp bend	55
90°, r/d=1	19
90°, r/d=1.5	13
90°, r/d=2	11
45°, sharp bend	18
45°, r/d=1	14
45°, r/d=1.5	9.4
Valves / Strainers	
Globe valve, full open	320
Gate valve, full open	7.5
Ball valve, full bore	2.6
Ball valve, reduced bore	25
Plug valve, 2-way	17
Plug valve, 3-way,straight thru	29
Plug valve, 3-way, branch	84

Fittings	L_e/D
Threaded tees	
Straight thru	20
Branch	60
Welded tees	
Square, straight thru	23
Square, branch	70
Radiused, straight thru	10
Radiused, branch	57
Valves / Strainers	
Diaphragm valve, weir type	160
Butterfly valve	37
Lift check valve	560
Swing check valve	95
Wafer disk check valve	420
Y-strainer, clean	250

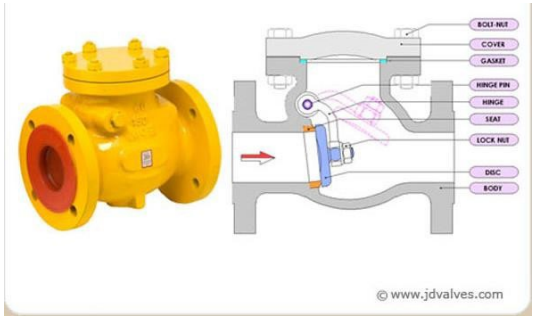
Valves and Fittings



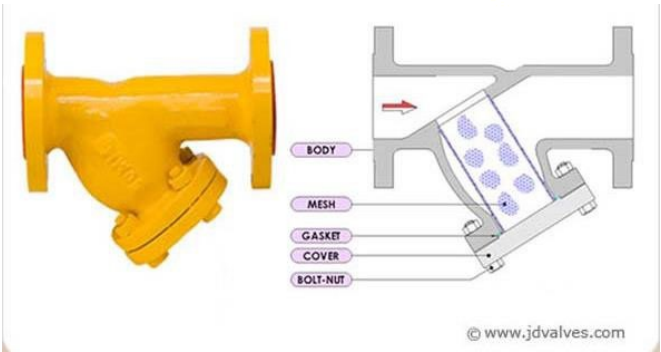


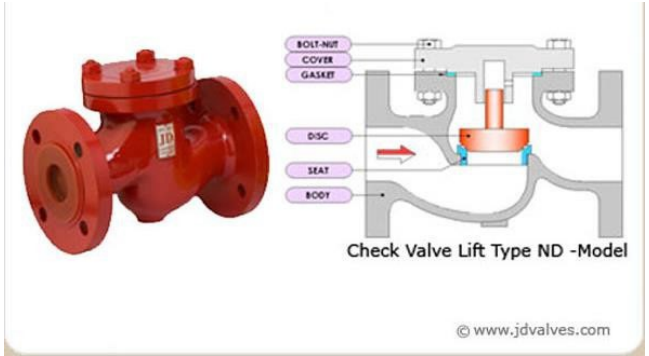










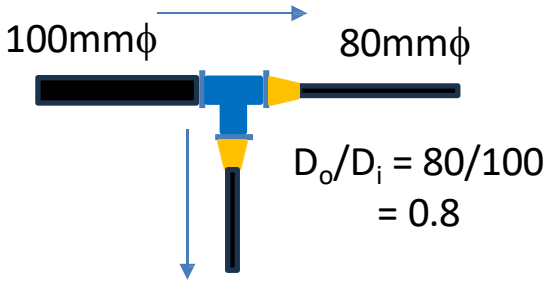


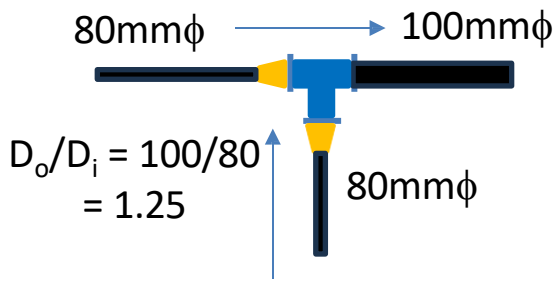
(Le/D) Ratio for Reducers and Enlargers

D_o/D_i	Reducer		D_o/D_i	Enlarger
0.9	3		1.1	1.5
0.8	8		1.3	8.5
0.7	18		1.5	16
0.6	38		1.7	22
0.5	85		2.0	28
0.4	220		2.5	35
			3.0	40
			4.0	44

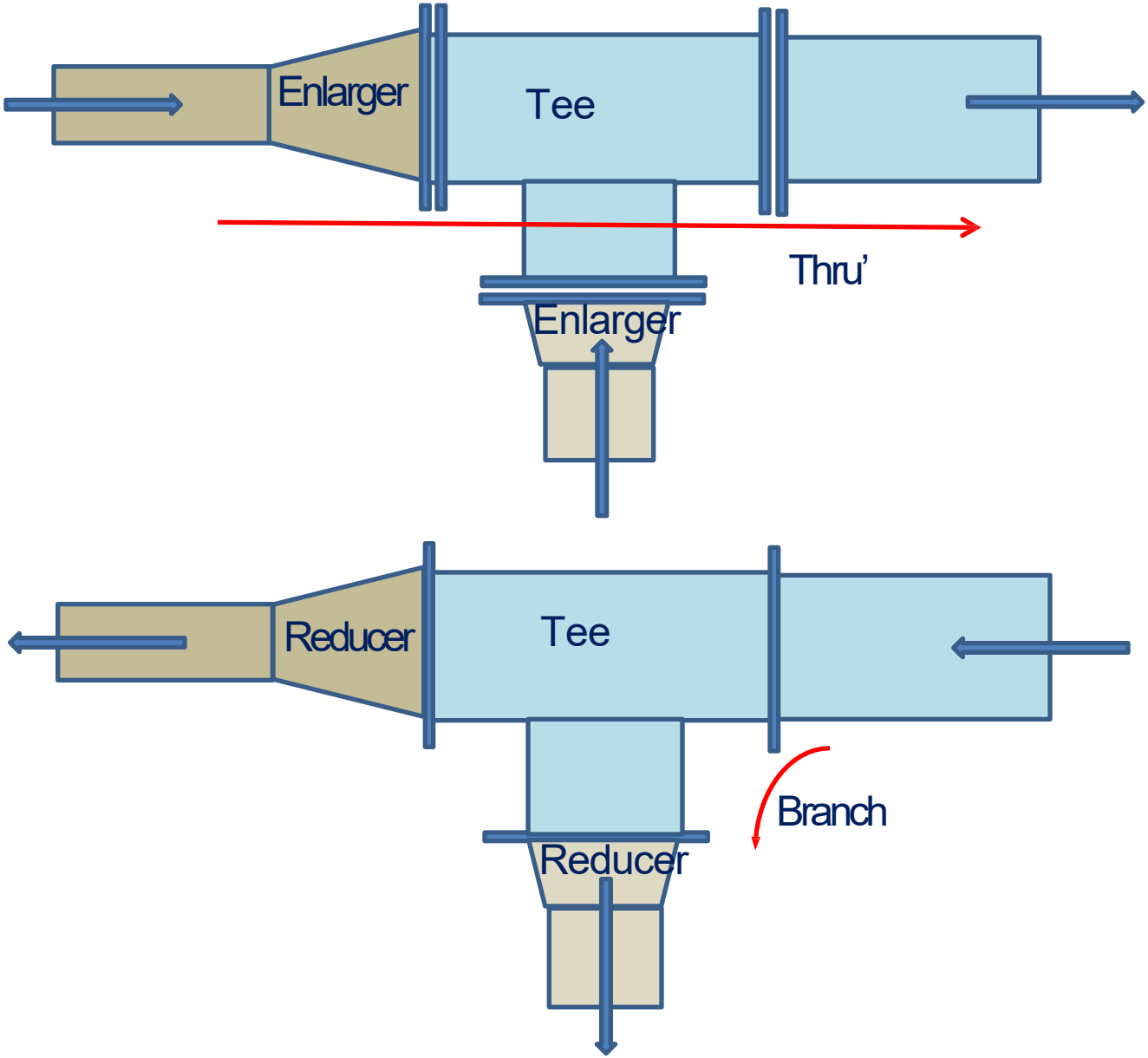
*Based on Upstream Diameter

(D_o/D_i = downstream diameter/upstream diameter)

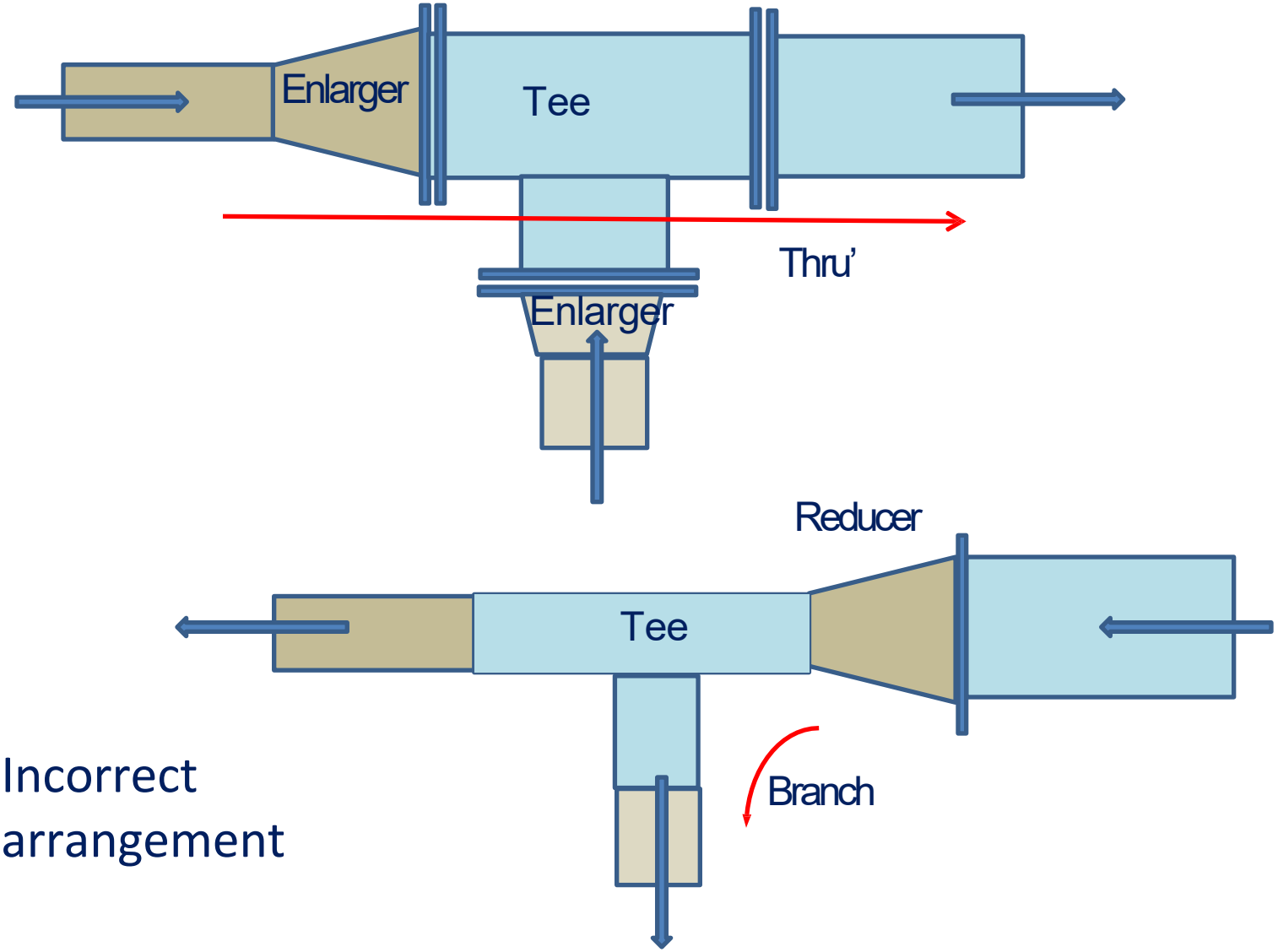




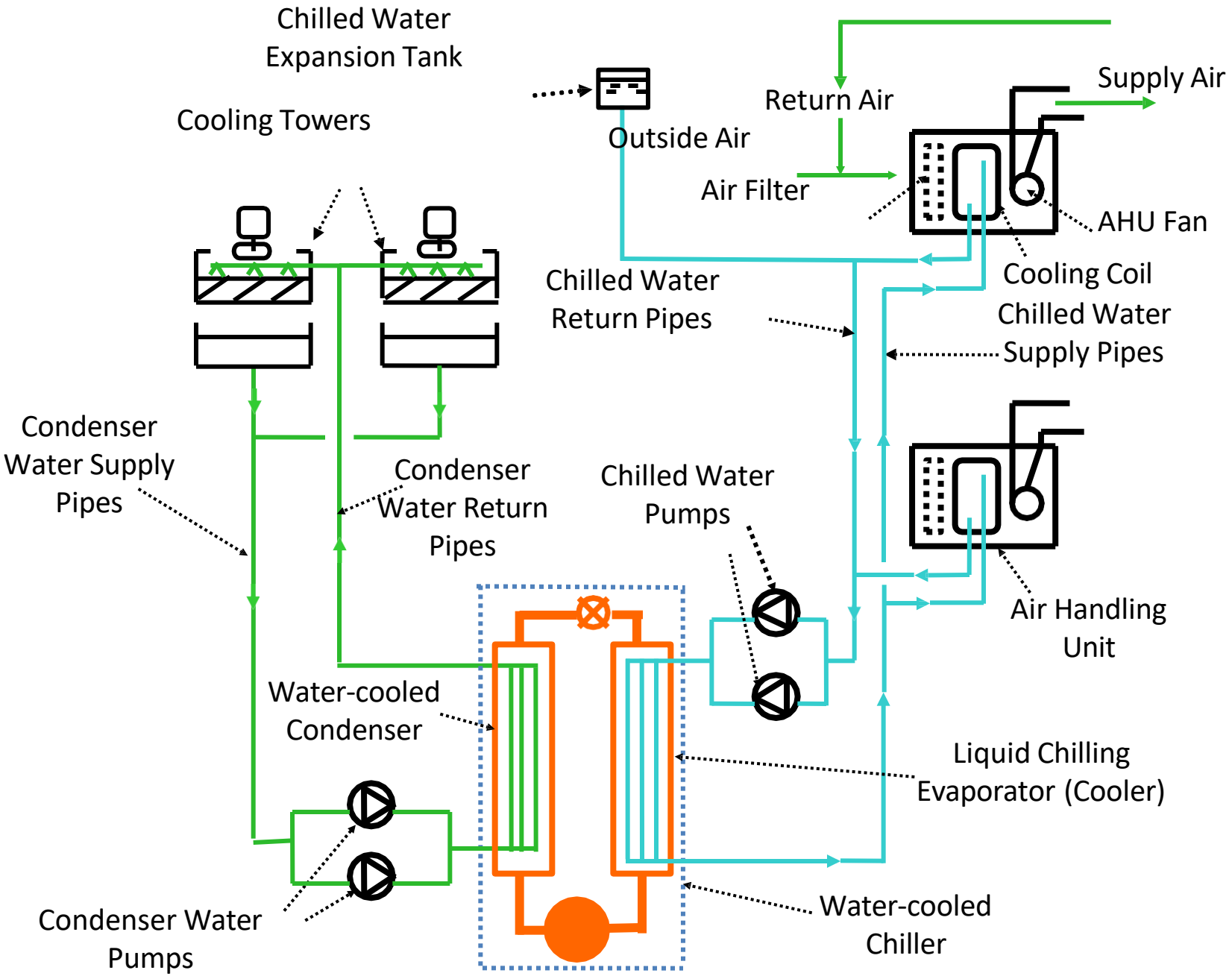
Pipe Sizing _ Guide



Pipe Sizing _ Guide



Incorrect arrangement


Chilled Water
Expansion Tank
Cooling Towers
Outside Air
Return Air
Supply Air
Air Filter
AHU Fan
Cooling Coil
Chilled Water
Supply Pipes
Chilled Water
Return Pipes
Condenser
Water Supply
Pipes
Condenser
Water Return
Pipes
Chilled Water
Pumps
Air Handling
Unit
Water-cooled
Condenser
Liquid Chilling
Evaporator (Cooler)
Condenser Water
Pumps
Water-cooled
Chiller

Example 7.1

Figure 7.12 shows the schematic diagram of the chilled water circuit of an air-conditioning system. One chilled water pump is on standby. The chilled water pumps are identical. Identify the index run and size the index run pipe sections using the equivalent length method and determine the required duty of each chilled water pump i.e. volume flow rate and head.

Example 7.1

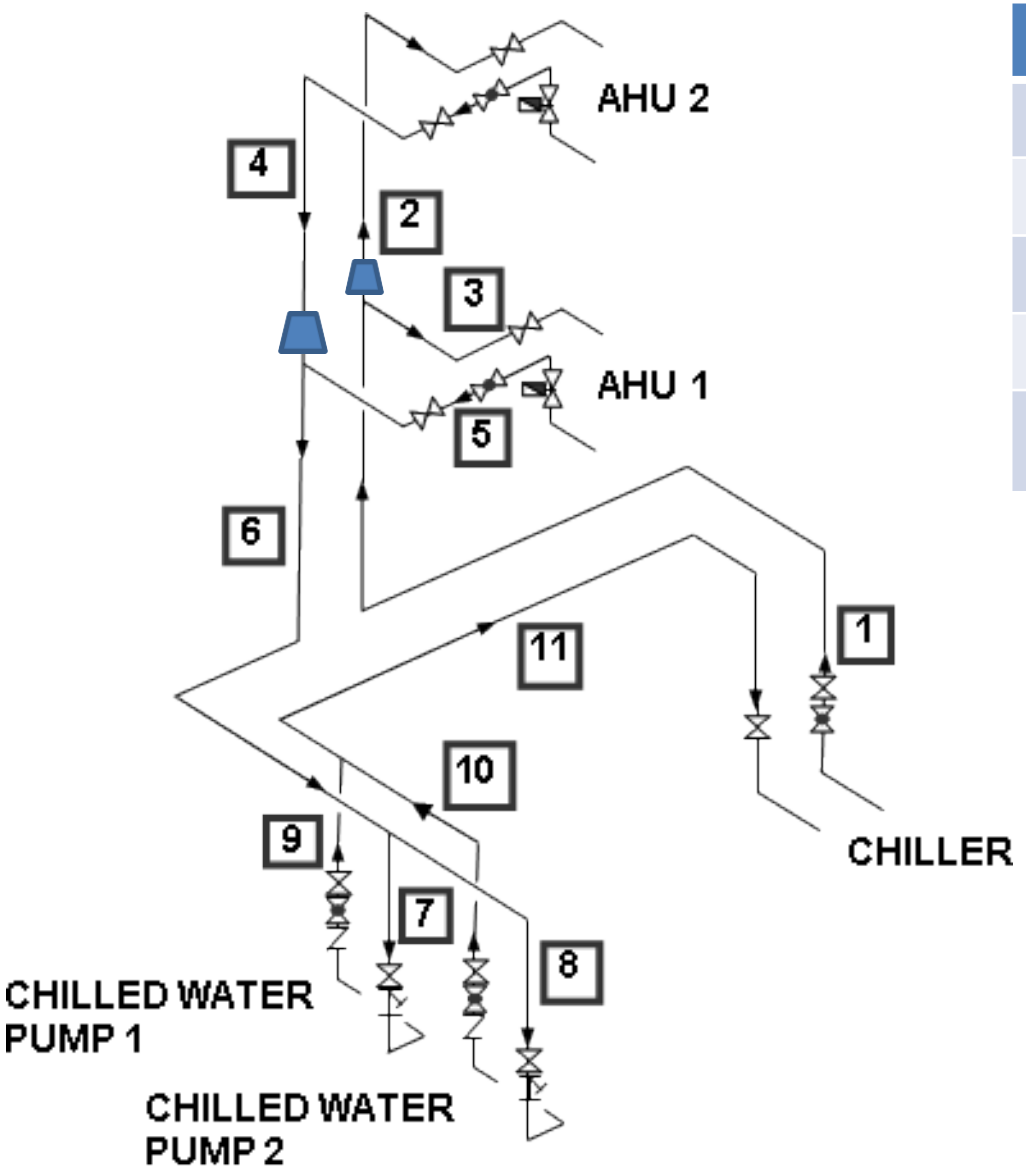


Figure 6.12

Legend		
1	Gate Valve	
2	Globe valve	
3	Check valve (Swing)	
4	Strainer (Y type)	
5	2-way modulating valve (globe)	

Section	Length (m)
1	20
2	10
3	4
4	10
5	4
6	15
7	3
8	5
9	3
10	5
11	10

Example

Data

Type of pipe material : Schedule 40 steel

Type of elbow : 90°, sharp bend, welded

Type of Tee : Square, welded

Equipment/Fittings	Flow rate (l/s)	Pressure Drop (kPa)
AHU 1	4	20
AHU 2	6	25
CHILLER	To be determined	50

Design Criteria

Range of acceptable pressure loss/m of pipe: 100 to 400 Pa/m

Example 7.1

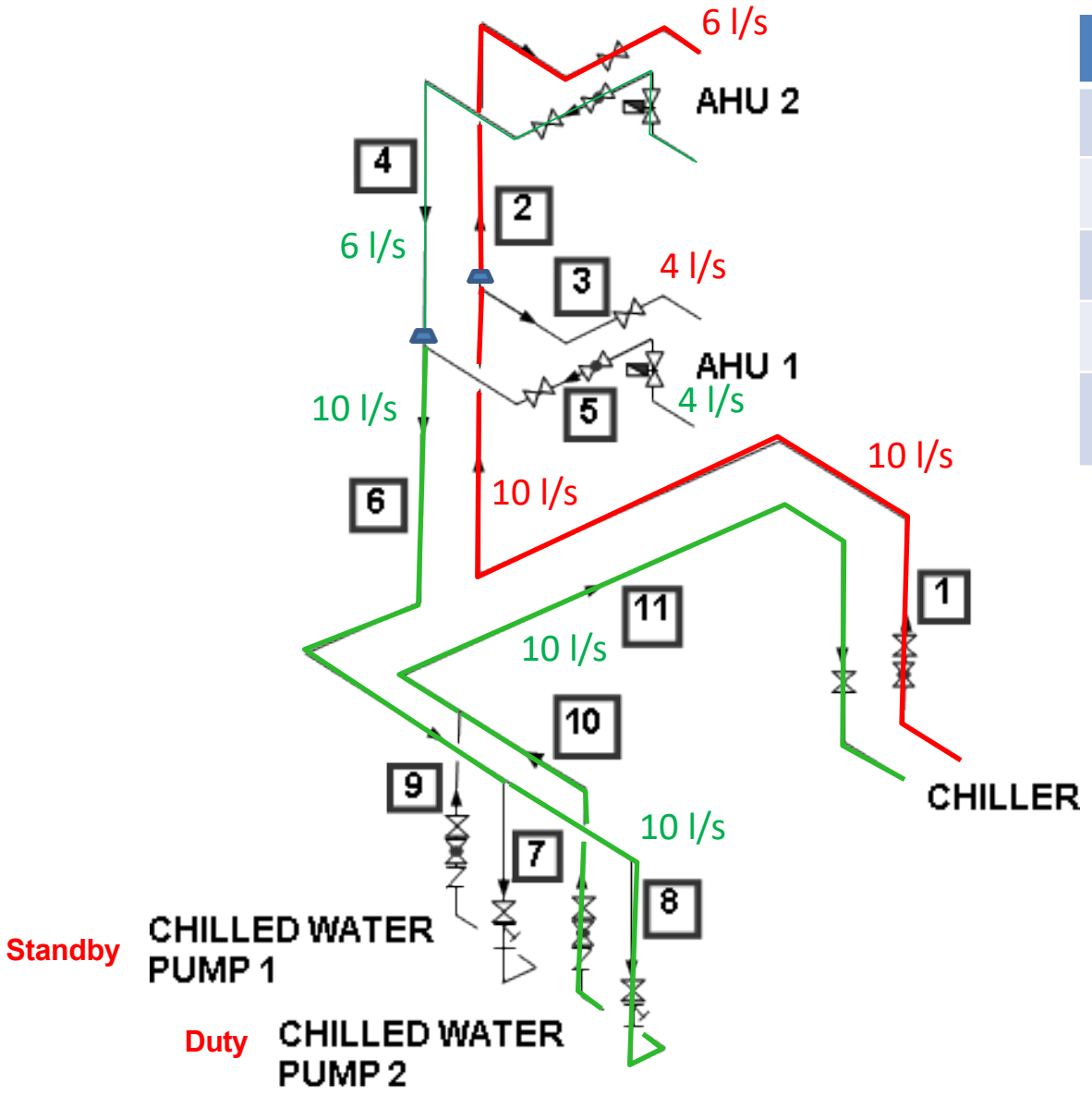


Figure 6.12

Legend		
1	Gate Valve	
2	Globe valve	
3	Check valve (Swing)	
4	Strainer (Y type)	
5	2-way modulating valve (globe)	

Section	Length (m)
1	20
2	10
3	4
4	10
5	4
6	15
7	3
8	5
9	3
10	5
11	10

Example 7.1

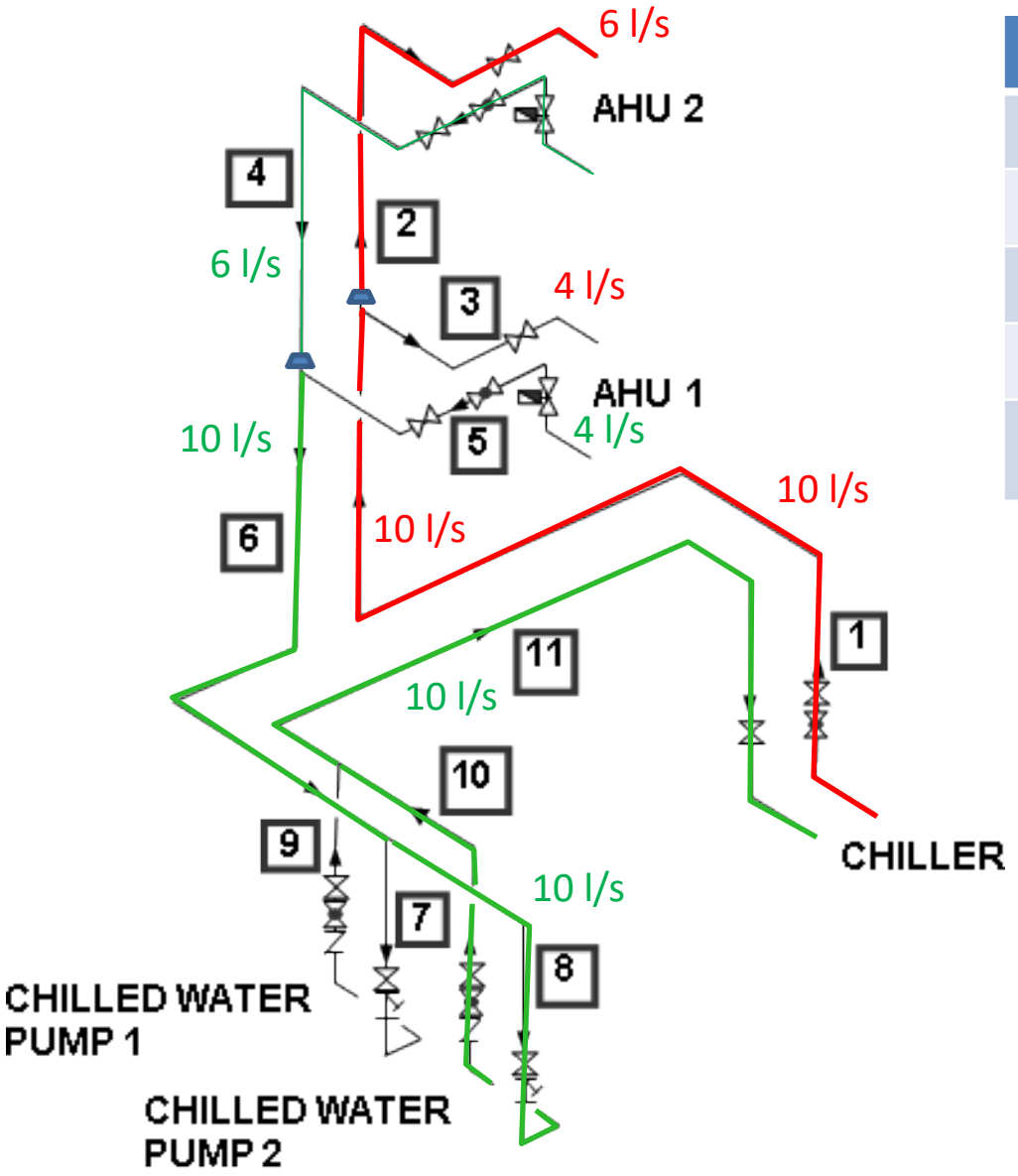
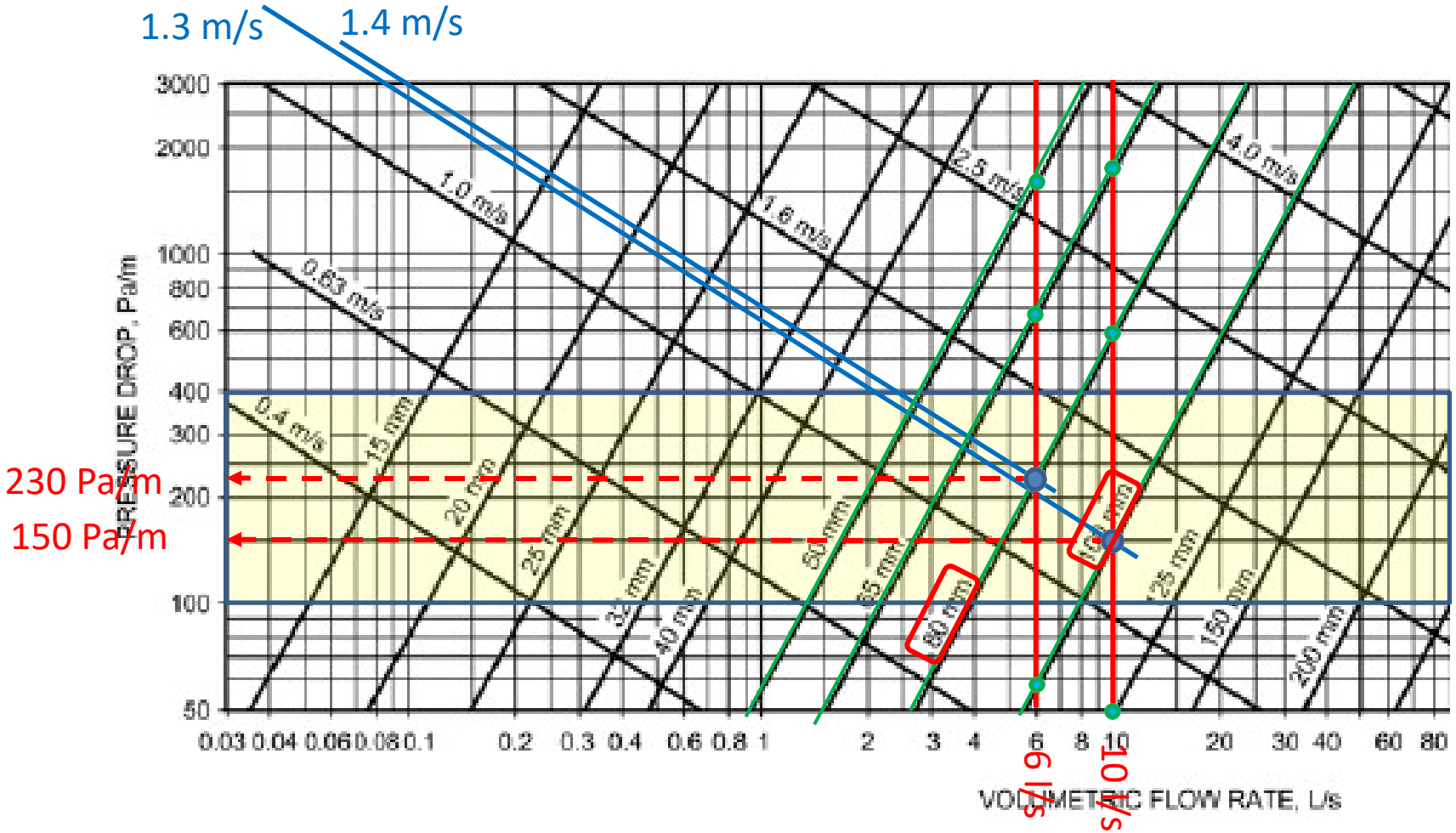


Figure 6.12

Legend		
1	Gate Valve	
2	Globe valve	
3	Check valve (Swing)	
4	Strainer (Y type)	
5	2-way modulating valve (globe)	

Section	Length (m)
1	20
2	10
3	4
4	10
5	4
6	15
7	3
8	5
9	3
10	5
11	10


1.3 m/s
1.4 m/s
230 Pa/m
150 Pa/m
6 l/s
10 l/s
PRESSURE DROP, Pa/m
VOLUMETRIC FLOW RATE, L/s
3000
2000
1000
800
600
400
300
200
100
50
0.03 0.04 0.06 0.08 0.1 0.2 0.3 0.4 0.6 0.8 1 2 3 4 6 8 10 20 30 40 60 80
0.4 m/s
0.63 m/s
1.0 m/s
1.6 m/s
2.5 m/s
4.0 m/s
15 mm
20 mm
25 mm
32 mm
40 mm
50 mm
65 mm
80 mm
100 mm
125 mm
150 mm
200 mm

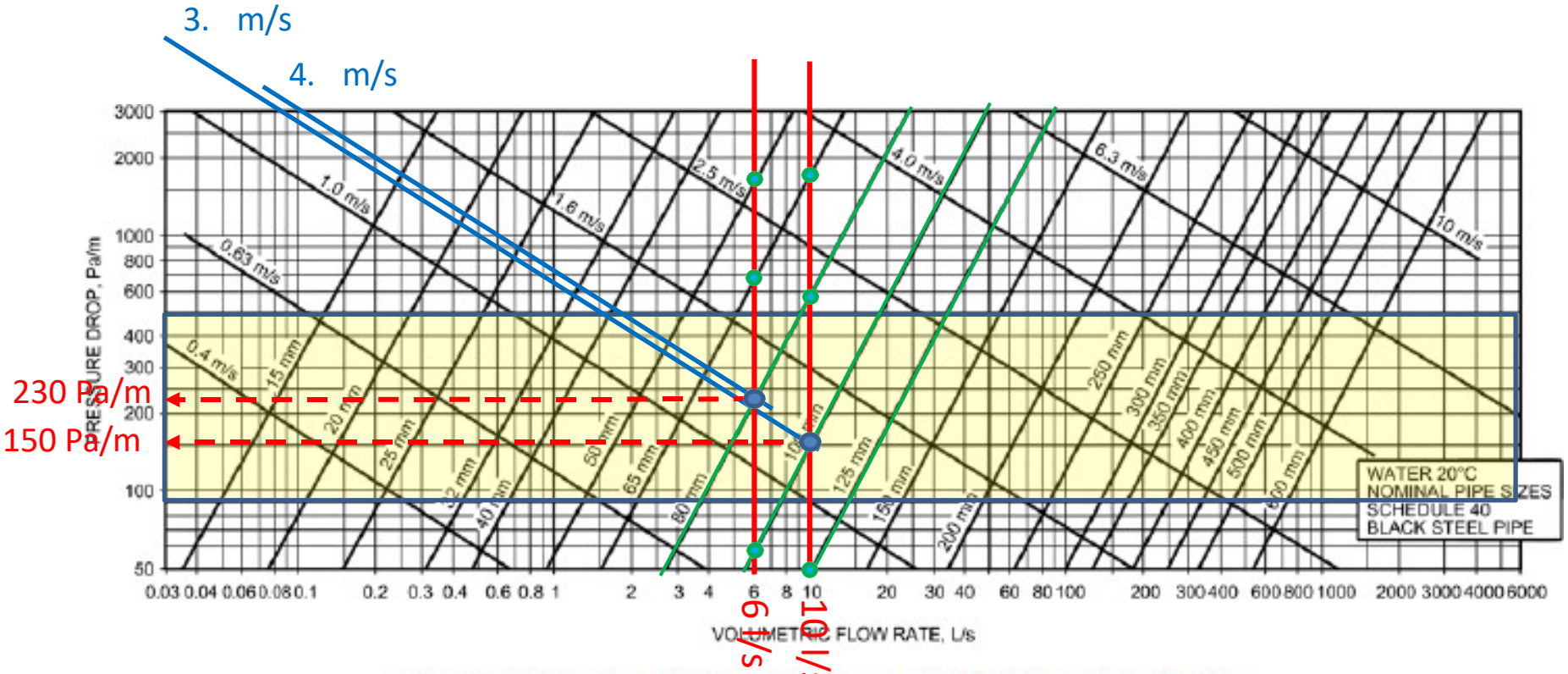


Fig. 4 Friction Loss for Water in Commercial Steel Pipe (Schedule 40)

Pipe Sizing Table (Equivalent Length)
Chilled Water Circuit

Index run is 1-2- 4 - AHU 2-6-8-10-11- Chiller Evaporator

Section	Volume Flow Rate (L/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
Sub-total carried forward from previous page									
1	10	100	1.3	Pipe	-	20	77.05	150	77.05 x 150
				Elbow	4	4 x 55 x 0.1			= 11,557.5
				Gate V	1	1 x 7.5 x 0.1			
				Globe V	1	1 x 320 x 0.1			
				Tee (T)	1	1 x 23 x 0.1			
2, 4	6	80	1.4	Pipe	-	10 + 10	104.52	230	104.52 x 230
				Elbow	7	7 x 55 x 0.08			= 24,039.6
				Gate V	2	2 x 7.5 x 0.08			
				Globe V	1	1 x 320 x 0.08			
				Modulating V	1	1 x 320 x 0.08			
				Reducer	1	1 x 8 x 0.08			
				Enlarger	1	1 x 8.5 x 0.08			
AHU 2									25,000
6, 8, 10, 11	10	100	1.3	Pipe	-	15 + 5 + 5 + 10	166.55	150	24,982.5
				Elbow	11	11 x 55 x 0.1			
				Gate V	3	3 x 7.5 x 0.1			
				Globe V	1	1 x 320 x 0.1			
				Strainer	1	1 x 250 x 0.1			
				Check V	1	1 x 95 x 0.1			
				Tee (T)	1	1 x 23 x 0.1			
Chiller									50,000
Sub-total to be carried forward									**135,579.6**

Pipe Sizing Table (Equivalent Length)
Chilled Water Circuit
Index run is 1-2-4-AHU 2-6-8-10-11- Chiller Evaporator

Section	Volume Flow Rate (L/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
Sub-total carried forward from previous page									
1	10	100	1.3	Pipe	-	20	77.05	150	77.05 x 150
				Elbow	4	4 x 55 x 0.1			= 11,557.5
				Gate V	1	1 x 7.5 x 0.1			
				Globe V	1	1 x 320 x 0.1			
				Tee (T)	1	1 x 23 x 0.1			
2, 4	6	80	1.4	Pipe	-	10 + 10	104.52	230	104.52 x 230
				Elbow	7	7 x 55 x 0.1			= 24,039.6
				Gate V	2	2 x 7.5 x 0.1			
				Globe V	1	1 x 320 x 0.1			
				Modulating V	1	1 x 320 x 0.1			
				Reducer	1	1 x 8 x 0.1			
				Enlarger	1	1 x 8.5 x 0.1			
AHU 2									25,000
6,8,10, 11	10	100	1.3	Pipe	-	15 + 5 + 5 + 10	166.55	150	24,982.5
				Elbow	11	11 x 55 x 0.1			
				Gate V	3	3 x 7.5 x 0.1			
				Globe V	1	1 x 320 x 0.1			
				Strainer	1	1 x 250 x 0.1			
				Check V	1	1 x 95 x 0.1			
				Tee (T)	1	1 x 23 x 0.1			
Chiller									50,000
Sub-total to be carried forward									135,579.6

Question Data Schematic **Pipe Sizing Table (Equivalent Length)** Le/D Le/D2 Chart

Chilled Water Circuit

Index run is 1-2-4-AHU 2-6-8-10-11- Chiller Evaporator

Section	Volume Flow Rate (L/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
							Sub-total carried forward from previous page		**135,579.6**
							Total Pressure Loss		135,579.6
							Safety Factor (10%)		13,558
							Grand Total Pressure Loss		135,579.6+ 13,558 = 149,137.6
							Frictional Head (m)		15.2
							Static Head (m)		0
							Pump Head (m)		15.2

Frictional Head Calculation:

Frictional head, $h_f = \frac{P}{\rho x g} = \frac{149,137.6}{1000 \times 9.81} = 15.2m$

Pipe Sizing Table (Equivalent Length)
Chilled Water Circuit
Index run is 1-2-4-AHU 2-6-8-10-11- Chiller Evaporator

Section	Volume Flow Rate (L/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
							Sub-total carried forward from previous page		135,579.6
								Total Pressure Loss	135,579.6
								Safety Factor (10%)	13,558
								Grand Total Pressure Loss	135,579.6+13,558 = 149,137.6
								Frictional Head (m)	15.2
								Static Head (m)	0
								Pump Head (m)	15.2

Frictional Head Calculation:

Frictional head, $h_f = \frac{P}{\rho_w \times g} = \frac{149,137.6}{1000 \times 9.81} = 15.2m$

Example

Figure shows the schematic diagram of the condenser water circuit of an air-conditioning system. All the equipment is in operation normally. The volume flow rate of condenser water through each chiller, cooling tower and condenser water pump is 10 l/s. Size the index run pipe sections using the Friction Loss chart. And hence determine the required condenser water pump head using the equivalent length method. Assume a 10% safety factor.

Example

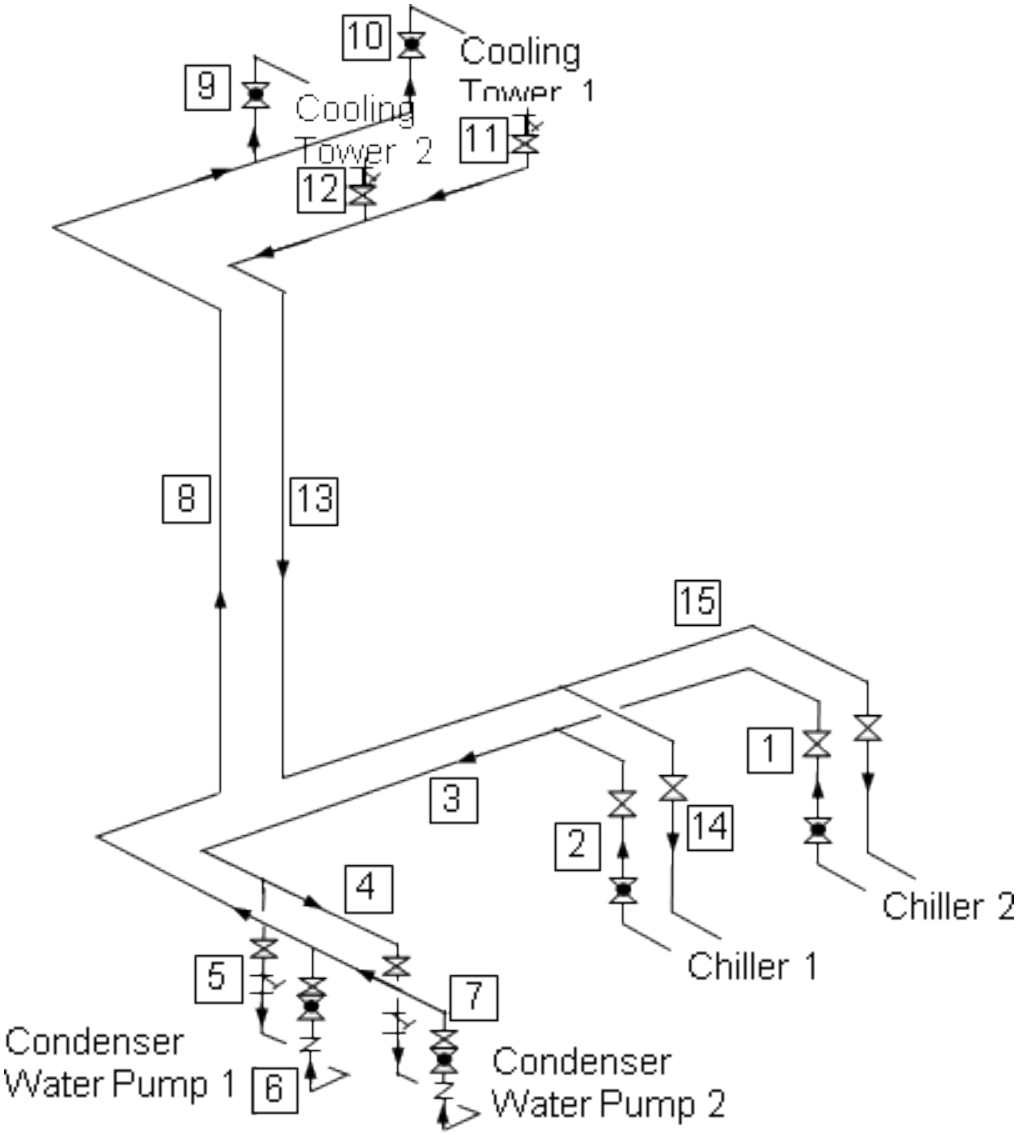


Figure 6.13

Legend	
	Gate Valve
	Globe valve
	Check valve (Wafer disk)
	Strainer (Y type)

Section	Length (m)
1	8
2	6
3	4
4	5
5	3
6	3
7	5
8	30
9	4
10	8
11	5
12	1
13	30
14	7
15	9

Example

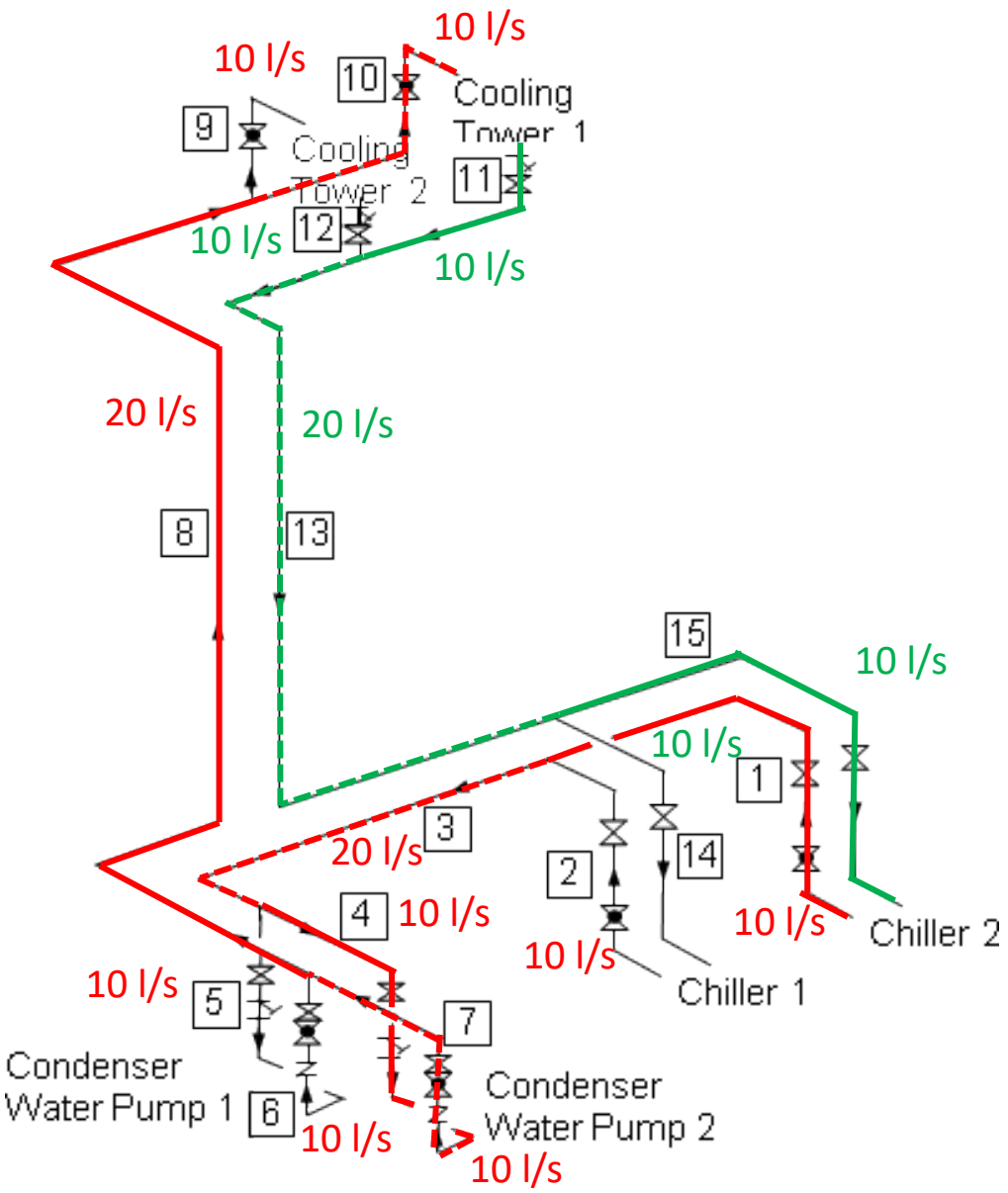


Figure 6.13

Legend	
	Gate Valve
	Globe valve
	Check valve (Wafer Disk)
	Strainer (Y type)

Section	Length (m)
1	8
2	6
3	4
4	5
5	3
6	3
7	5
8	30
9	4
10	8
11	5
12	1
13	30
14	7
15	9

Pipe Sizing Table (Equivalent Length)
Condenser Water Circuit

Question Data Schematic Index run is 1-3-4-7-8-10-11-13-15-Chiller Condenser Le/D Le/D2 Chart

Section	Volume Flow Rate (L/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
Sub-total carried forward from previous page									
1	10	100	1.3	Pipe	-	8	44.95	150	44.95 x 150
				Elbow	3	3x 14 x 0.1			= 6,742.5
				Gate V	1	1 x 7.5 x 0.1			
				Globe V	1	1 x 320 x 0.1			
3	20	125	1.55	Pipe	-	4	11.5	180	11.5 x 180
				Elbow	1	1 x 14 x 0.125			= 2,070
				Tee(T)	2	2 x 23 x 0.125			
4,7	10	100	1.3	Pipe	-	5 + 5	117.5	150	17,625
				Elbow	5	5x 14 x 0.1			
				Gate V	2	2 x 7.5 x 0.1			
				Strainer	1	1 x 250 x 0.1			
				Globe V	1	1 x 320 x 0.1			
				Check V	1	1 x 420 x 0.1			
8	20	125	1.55	Pipe	-	30	42.75	180	7,695
				Elbow	4	4 x 14 x 0.125			
				Tee(T)	2	2 x 23 x 0.125			

Question Data Schematic

Pipe Sizing Table (Equivalent Length)
Condenser Water Circuit

Le/D Le/D2 Chart

Index run is 1-3-4-7-8-10-11-13-15-Chiller Condenser

Section	Volume Flow Rate (L/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
10,11	10	100	1.3	Pipe	-	8 + 5	74.95	150	11,242.5
				Elbow	3	3 x 14x 0.1			
				Gate V	1	1 x 7.5 x 0.1			
				Globe V	1	1 x 320 x 0.1			
				Strainer	1	1 x 250 x 0.1			
								Sub-total to be carried forward	45,375

Question Data Schematic

Pipe Sizing Table (Equivalent Length)
Condenser Water Circuit

Le/D Le/D2 Chart

Index run is 1-3-4-7-8-10-11-13-15-Chiller Condenser

Section	Volume Flow Rate (L/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
						Sub-total carried forward from previous page			45,375
13	20	125	1.55	Pipe	-	30	41	180	41 x 180
				Elbow	3	3 x 14 x 0.125			= 7,380
				Tee(T)	2	2 x 23 x 0.125			
15	10	100	1.3	Pipe	-	9	13.95	150	13.95 x 150
				Elbow	3	3x 14 x 0.1			= 2,092.5
				Gate V	1	1 x 7.5 x 0.1			
Chiller									30,000
								Total Pressure Loss	84.847.5
								Safety Factor (10%)	8484.8
								Grand Total Pressure Loss	84,847.5+8484.8 = 93,332.3
								Frictional Head (m)	9.51
								Static Head (m)	2
								Pump Head (m)	11.51

Frictional Head Calculation:

Frictional head, $h_f = \frac{P}{\rho_w \times g} = \frac{93,332.3}{1000 \times 9.81} = 9.51$ m